

7

NUMBER AND ALGEBRA

GRAPHING CURVES

When an object is thrown upwards, its path is a curve called a parabola. The shape and length of the path will depend on the initial speed of the object. Furthermore, car headlights and satellite dishes all use mirrors or reflectors that have the shape of a parabola.



Chapter outline

		Working mathematically				
7.01	Direct variation*	U	F	PS	R	C
7.02	Inverse variation*	U	F	PS	R	C
7.03	Conversion graphs*	U	F	PS	R	C
7.04	Distance–time graphs*	U	F		R	C
7.05	Graphs of variable change*	U	F		R	C
7.06	Graphing quadratic equations $y = ax^2 + c$	U	F		R	C
7.07	Graphing quadratic equations $y = a(x - b)^2 + c$*	U	F		R	
7.08	Applying quadratic equations	U	F	PS	R	C
7.09	Graphing cubic equations $y = ax^3 + c$*	U	F		R	C
7.10	Graphing power equations $y = ax^n + c$*	U	F		R	C
7.11	Graphing hyperbolas $y = \frac{k}{x} + c$*	U	F		R	C
7.12	Graphing exponential equations $y = ka^x + c$	U	F		R	C
7.13	Exponential growth and decay	U	F	PS	R	C
7.14	Graphing logarithmic equations $y = \log_a x$*	U	F		R	C
7.15	Graphing circles $(x - a)^2 + (y - b)^2 = r^2$*	U	F		R	C
7.16	Identifying graphs	U	F		R	C

***PATH**

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communication

Wordbank

asymptote A line that a curve gets very close to but never touches, for example, the x-axis is an asymptote of the exponential curve

direct variation or **direct proportion** A relationship between 2 variables of the form $y = kx$, where k is a constant; for example, if $y = 8.5x$, then y is directly proportional to x

exponential equation An equation involving a variable as a power, such as $y = 3^x$, whose graph is an exponential curve

hyperbola The graph of $y = \frac{k}{x}$, where k is a constant, which has 2 branches and 2 asymptotes

inverse variation or inverse proportion A relationship between 2 variables of the form $y = \frac{k}{x}$, where k is a constant; for example, if $y = \frac{50}{x}$, then y is inversely proportional to x

parabola A U-shaped curve that is the graph of a quadratic equation

quadratic equation An equation involving a variable squared (power of 2), such as $y = 3x^2 - 6$, whose graph is a curve called a parabola

Quiz
Wordbank 7

Videos (11):

7.01 Direct proportion • Direct linear variation
7.02 Inverse variation • Inverse proportion
7.05 Graphs of variable change
7.06 The parabola $y = ax^2 + c$
7.07 Graphing the parabola $y = a(x - b)^2 + c$
7.08 Applying quadratic functions
7.11 Graphing hyperbolas
7.12 The exponential curve
7.16 Identifying graphs

***Twig* videos (2):**

7.02 The heartbeat formula
7.12 The emperor's chessboard

***PhET* interactive (1):**

7.06 Graphing quadratics

Quizzes (5):

- Wordbank 7
- SkillCheck 7
- Mental skills 7
- Language of maths 7
- Test yourself 7

Worksheets (11):

7.02 Direct and inverse proportion • Variation problems
7.03 Currency conversion graph
7.05 Graphs of change
7.06 Graphing parabolas
7.07 Matching parabolas • Matching parabolas with their equations
7.09 Graphing cubics 1
7.11 Graphing hyperbolas
7.12 Graphing exponentials
Mind map Graphing curves (Advanced)

Puzzles (5):

7.04 Jane's diary
7.15 Circle equations
7.16 Matching graphs • Matching graphs (Advanced)
Language of maths Graphing curves crossword

Technology (4):

7.06 Investigating parabolas 1 • Curve sketcher
7.07 Investigating parabolas 2
7.10 Power functions
7.11 Curve sketcher

Spreadsheets (4):

7.06 Investigating parabolas 1 • Curve sketcher
7.07 Investigating parabolas 2
7.10 Power functions
7.11 Curve sketcher

Nelson MindTap

To access resources above, visit **cengage.com.au/nelsonmindtap**

In this chapter you will:

- (PATH) solve problems involving direct variation and inverse variation
- (PATH) read and interpret conversion graphs, distance–time (travel) graphs where the speed is variable, and graphs of variables changing at different rates
- graph parabolas of the form $y = ax^2 + c$ and (PATH) $y = a(x - b)^2 + c$
- (PATH) graph cubic curves of the form $y = ax^3 + c$
- (PATH) graph higher-power curves of the form $y = ax^n + c$ and $y = a(x - b)^n$
- (PATH) graph hyperbolas of the form $y = \frac{k}{x} + c$ and $y = \frac{k}{x - b}$
- graph exponential curves of the form $y = a^x + c$
- (PATH) graph exponential curves of the form $y = ka^x + c$
- solve problems that can be modelled by quadratic and exponential equations
- (PATH) graph logarithmic curves of the form $y = \log_a x$
- (PATH) graph circles of the form $x^2 + y^2 = r^2$ and $(x - a)^2 + (y - b)^2 = r^2$
- match graphs to their equations

SkillCheck ANSWERS ON P. 635

1 If $A = 2x^2 - 3$, find A if:

a $x = 1$ **b** $x = 4$ **c** $x = 0$ **d** $x = -6$

2 If $y = 5^x$, find y if:

a $x = 4$ **b** $x = 5$ **c** $x = 0$ **d** $x = -2$

3 If $S = \frac{120}{t}$, find S if:

a $t = 10$ **b** $t = 35$ **c** $t = 2.5$ **d** $t = 1.8$

Quiz
SkillCheck 7

Direct variation 7.01

PATH

Videos
Direct proportion
Direct linear variation

Two variables are **directly proportional** to each other if one variable is a constant multiple of the other; when one variable changes, the other one changes by the same factor.
This relationship is called **direct variation** or **direct proportion**.

Direct variation

If y is **directly proportional** to x, written as '$y \propto x$', then $y = kx$, where k is a constant (number) called the **constant of variation** or **constant of proportionality**.

- A direct linear relationship exists between x and y
- If x increases (or decreases), y increases (or decreases)
- If x is doubled (or halved), y is doubled (or halved)
- Another way of saying 'y is directly proportional to x' is 'y **varies directly** as x'
- The graph of direct variation is a straight line going through (0, 0) with gradient k

Example 1

The distance (d) in metres travelled by a cyclist on a road bike is directly proportional to the number of rotations (r) of its wheels. After 750 rotations, a distance of 1600 m is travelled.

a What distance will be travelled after 1200 rotations?

b How many full rotations will be needed to cover 30 km?

Solution

a d is directly proportional to r, or $d \propto r$.

$\therefore d = kr$

To find k, substitute the information given for r and d.

When $r = 750$, $d = 1600$:

$1600 = k \times 750$

$k = \frac{1600}{750}$

$k = 2.133...$

PATH

$\therefore d = 2.133\ldots r$

When $r = 1200$,

$d = 2.133\ldots \times 1200$

$= 2560$ m

After 1200 rotations, the distance travelled will be 2560 m.

b When $d = 30$ km $= 30\,000$ m — Distance must be in metres

$30\,000 = 2.133\ldots \times r$

$r = \frac{30\,000}{2.133\ldots}$

$= 14\,062.5$

$\approx 14\,063$ rotations — Rounding up for full rotations

For a distance of 30 km, there will be 14 063 rotations.

Solving direct variation problems

1 Identify the 2 variables (say x and y) and form a variation equation, $y = kx$
2 Substitute values for x and y to find k, the constant of variation
3 Rewrite $y = kx$ using the value of k
4 Substitute a value for x or y into $y = kx$ to solve the problem

Example 2

M varies directly as n. If when $n = 6$, $M = 103.8$, find M when $n = 14.2$.

Solution

$M = kn$

To find k, substitute $n = 6$ and $M = 103.8$.

$103.8 = k \times 6$

$6k = 103.8$

$k = \frac{103.8}{6}$

$= 17.3$

$\therefore M = 17.3n$

Substitute $n = 14.2$ to find M.

$M = 17.3 \times 14.2$

$= 245.66$

EXERCISE 7.01 ANSWERS ON P. 635

PATH

Direct variation U F PS R C

EXAMPLE 1

7.01

1 The distance, D, travelled by Connor, a marathon runner, varies directly as time, T. PS R C

Time, T (min)	1	2	3
Distance, D (m)	190	380	570

a Find the constant of variation and write the equation for D.

b How far, in kilometres, will Connor run in:

i 20 minutes? **ii** 45 minutes?

c How long would it take Connor to run 12.35 kilometres? Answer in hours and minutes.

2 Mehta's earnings for working a shift at the local nursery are directly proportional to the number of hours she works. Yesterday, she earned \$304.30 for working an 8.5 hour shift. PS R C



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a If Mehta's earnings are represented by E, and the number of hours worked is represented by h, find the constant of variation and write an equation for E.

b How much will she earn for working a 7-hour shift?

c How many hours did she work today if she earned \$196.90 for the shift?

3 The amount of interest, I, earned for one year on an investment account varies directly as the size of the deposit, D. PS R C

a If Caterina earns \$826.50 interest on an investment of \$14 250, find the variation equation for I, including the constant of variation.

b Hence, how much will she earn on an investment of \$9000?

c If she doubles the size of her investment in **b**, how much will she earn in interest?

EXAMPLE 2

4 S varies directly as t. If when $t = 14$, $S = 106.4$, what is the value of S when $t = 0.3$? Select the correct answer **A**, **B**, **C** or **D**. R C

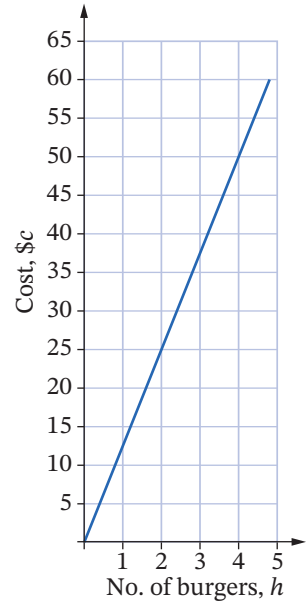
A 2.28 **B** 27.72 **C** 36.12 **D** 446.88

5 Find the linear formula for b in terms of a for this table of values.

a	4	8	12	16	20
b	10	20	30	40	50

PATH

6 The line graph shows that the cost of hamburgers purchased from the local takeaway store depends directly on the number of burgers purchased. PS R C



a Copy this table and use the graph to complete it.

No. of burgers, h	1	2	3
Cost, c ($)			

b Find the variation equation to represent the relationship between the cost ($\$c$) and the number of burgers (h).

c If Kendall buys 6 hamburgers, what is the total cost of the hamburgers?

d The total cost of one order of hamburgers is $137.50. How many hamburgers were ordered?

e Find the gradient of the line. How is it related to the constant of variation?

7 K varies directly as L. If $L = 9.5$ when $K = 1045$, what is the value of K when $L = 1.65$? Select **A**, **B**, **C** or **D**. R C

A 0.015 **B** 93.7 **C** 181.5 **D** 1708.575

8 A linear relationship exists between the mass of a car (m kg) and its rate of fuel consumption (F L/100 km). PS R C

a Find the variation equation for F if a 1500 kg car uses fuel at a rate of 9 L/100 km.

b Find the fuel consumption of a 2500 kg car.

9 For an object that is cooling, the drop in temperature varies directly with time. If the temperature drops 8°C in 5 minutes, how long would it take to drop 10°C? Select **A**, **B**, **C** or **D**. PS R C

A 6.25 min **B** 7 min **C** 12.8 min **D** 16 min

10 The weight of an astronaut on Mars is proportional to his weight on Earth. A 72 kg astronaut weighs 27.4 kg on Mars. PS R C

a Calculate how much a 60 kg astronaut weighs on Mars, correct to 1 decimal place.

b If an astronaut weighs 32 kg on Mars, calculate his weight on Earth, correct to 1 decimal place.



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Inverse variation

7.02

Two variables are **inversely proportional** to each other if when one variable increases, the other one decreases by the same factor. This relationship is called **inverse variation** or **inverse proportion**.

The table below shows the different speeds of a car (s km/h), and the time it takes to travel 100 km (t min). As the speed increases, the time taken decreases.

Speed (s km/h)	50	60	80	100
Time (t min)	120	100	75	60

Inverse variation

If y is **inversely proportional** to x, written as '$y \propto \frac{1}{x}$', then $y = \frac{k}{x}$, where k is a constant (number) called the **constant of variation** or **constant of proportionality**.

- If x increases, y decreases ('inverse' means 'opposite')
- If x decreases, y increases
- If x is doubled, y is halved
- If x is halved, y is doubled
- Another way of saying 'y is inversely proportional to x' is 'y **varies inversely** as x'

Example 3

The time (t) in minutes taken by a car to travel 100 km is inversely proportional to the speed (s km/h) of the car, as shown in the table above. At 50 km/h, the time taken is 120 minutes.

a Find the constant of variation and the inverse variation equation for t.

b How long did the car take to travel 100 km at:

i 40 km/h? **ii** 110 km/h?

c Find the car's speed if it took 45 minutes to travel 100 km.

Solution

a t is inversely proportional to s.

$\therefore t = \frac{k}{s}$ Forming an inverse variation equation

To find k, substitute the information given for s and t.

When $s = 50$, $t = 120$: Finding k, the constant of variation

$120 = \frac{k}{50}$

$k = 120 \times 50$

$= 6000$

The constant of variation is 6000.

$\therefore t = \frac{6000}{s}$ Rewriting the equation $t = \frac{k}{s}$

PATH

Worksheets
Direct and inverse proportion
Variation problems

Videos
Inverse variation
The heartbeat formula

PATH

b **i** When $s = 40$, $t = \frac{6000}{40} = 150$ min

At 40 km/h, the trip takes 150 min (or 2 h 30 min).

ii When $s = 110$,

$t = \frac{6000}{110}$

$= 54.5454...$

≈ 55 min

At 110 km/h, the trip takes 55 min.

c When $t = 45$,

$45 = \frac{6000}{s}$

$45s = 6000$

$s = \frac{6000}{45}$

$= 133\frac{1}{3}$ km/h

For a travel time of 45 min, the speed must be $133\frac{1}{3}$ km/h.

Solving inverse variation problems

1 Identify the 2 variables (say x and y) and form a variation equation, $y = \frac{k}{x}$

2 Substitute values for x and y to find k, the constant of variation

3 Rewrite $y = \frac{k}{x}$ using the value of k

4 Substitute a value for x or y into $y = \frac{k}{x}$ to solve the problem

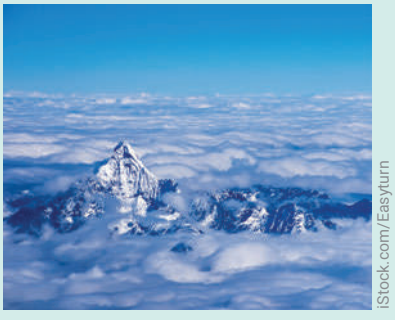
Video
Inverse proportion

Example 4

The temperature, T (in degrees Celsius), of the air is inversely proportional to the height, h (in metres), above sea level. At 600 m above sea level, the temperature is 8°C.

a What is the temperature at 1000 m above sea level?

b Graph the relationship between temperature and height above sea level, for heights between 0 and 5000.



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Solution

a T is inversely proportional to h.

$T = \frac{k}{h}$ — Forming a variation equation

Substitute $h = 600$ and $T = 8$ to find k.

$8 = \frac{k}{600}$ — Finding k

$k = 8 \times 600$

$= 4800$

$\therefore T = \frac{4800}{h}$ — Rewriting the equation $T = \frac{k}{h}$

When $h = 1000$, — Solving the problem

$T = \frac{4800}{1000} = 4.8$°C

The temperature at a height of 1000 metres above sea level is 4.8°C.

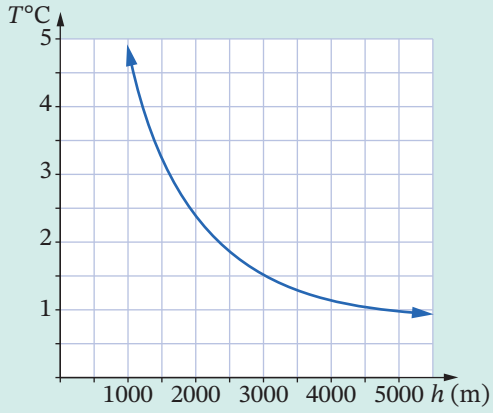
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b Draw a table of values for $T = \frac{4800}{h}$

h	1000	2000	3000	4000	5000
T	4.8	2.4	1.6	1.2	0.96

Note that as h increases, T decreases.

This graph is called a **hyperbola**.



PATH

7.02

EXERCISE 7.02 ANSWERS ON P. 635

Inverse variation U F PS R C

EXAMPLE 3

1 The time taken, T hours, to travel from Sydney to Melbourne varies inversely as its average speed, s km/h. PS R C

a If it takes 11.5 hours at an average speed of 80 km/h, find the constant of variation and the variation equation for T.

b If the average speed is increased to 90 km/h, how long will the journey take? Answer in hours and minutes.

c Find the average speed needed to complete the trip in 10 hours.

EXAMPLE 4

2 Air temperature, T (in degrees Celsius), varies inversely as its height, h (in metres), above sea level. At 150 m above sea level, the temperature is 30°C. PS R C

a Find the constant of variation and write a variation equation for T.

b What is the temperature at:

i 300 m above sea level? **ii** 2500 m above sea level?

c What is the height above sea level when the temperature is:

i 8°C? **ii** 22.5°C?

d Graph the relationship between temperature and height above sea level. Use T on the vertical axis and h on the horizontal axis with $h = 0, 500, 1000, 1500, ..., 3000$.

3 The maximum number of diners, N, in Yen's restaurant is inversely proportional to the amount of floor space, S m², allocated to each diner. If 1.4 m² is allowed per diner, the restaurant can have 80 diners. PS R C

a Write a variation equation for N.

b How many diners could the restaurant contain if only 1.1 m² was allocated per person?

c How much space (correct to one decimal place) is allocated to each diner if the restaurant has a maximum capacity of 55 diners?

d During the COVID-19 pandemic of 2020, social distancing measures introduced to minimise the spread of the virus allocated 4 m² to each diner. How many diners could Yen's restaurant serve under this restriction?

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4 The rate of vibration of a string varies inversely as its length. A string that is 330 mm long vibrates at 196 Hz (hertz). What length of string will vibrate at 500 Hz? Select the correct answer **A**, **B**, **C** or **D**. PS R C

A 129 mm **B** 297 mm **C** 323 mm **D** 842 mm

5 Which equation represents this table of values? Select **A**, **B**, **C** or **D**. R

x	2	5	8	10
y	2.5	1	0.625	0.5

A $y = \frac{10}{x}$ **B** $y = \frac{5}{x}$ **C** $y = \frac{2.5}{x}$ **D** $y = \frac{1}{x}$

6 K is inversely proportional to L. If $L = 2$ when $K = 7$, find K when $L = 15$. R

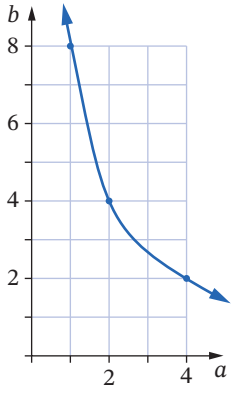
7 Ankit believes that at a train station, the number of people waiting on the platform is inversely proportional to the time until the next train arrives. According to his model, when there are 15 people waiting, the train will arrive in 4 minutes. PS R C

a When will the train arrive if there are 25 people waiting?

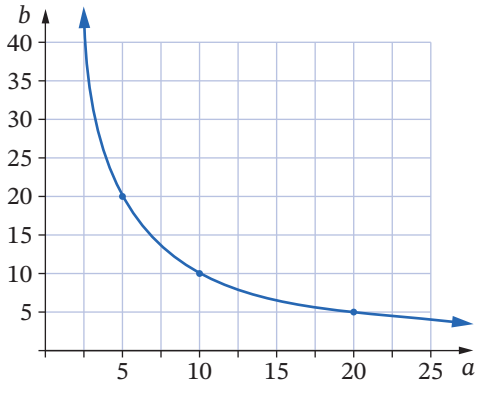
b How many people are waiting at the station 12 minutes before the train arrives?

8 Each graph shows an inverse relationship between a and b. Find each variation equation. R C

a



b



9 The time taken to travel from Sydney to Brisbane varies inversely as the average speed. A train travelling at an average speed of 70 km/h takes 14 h 7 min. How long would the same trip take if the average speed of the train was: PS R C

a 50 km/h? **b** 75 km/h?

10 The amount of time it takes Donia to paint her house is inversely proportional to the number of friends she has to help her. When she has 5 friends helping, the job takes 4 days. PS R C

a How long will it take if she has 3 friends helping?

b How many friends must she have to help her to paint the house in 2 days?

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11 The microwave oven cooking time (t seconds) for a meal varies inversely as the oven's power rating (W watts). For an 800 W microwave oven, the cooking time is 300 seconds. PS R C

a Find the variation equation for t.

b What is the cooking time for the meal in a 1200 W microwave oven?

c Find the power rating (nearest 100 W) of an oven if the cooking time is:

i 6 min 20 s **ii** 2 min 40 s

12 The pressure (P atmospheres) of an amount of gas varies inversely as its volume (V litres). For a particular gas, its pressure is 5.2 atm if its volume is 4.5 L. PS R C

a Find the variation equation for P.

b Calculate the volume of the gas when its pressure is 2.8 atm.

c If its volume increases to 8 L, what is its pressure?

PATH

7.03

Conversion graphs 7.03

A **conversion graph** is used to convert from one unit to another; for example, miles to kilometres, or Australian dollars to US dollars. It is usually a straight line that begins at (0, 0), so it has an equation $y = kx$ and is an example of **direct variation**.

PATH

Worksheet
Currency conversion graph

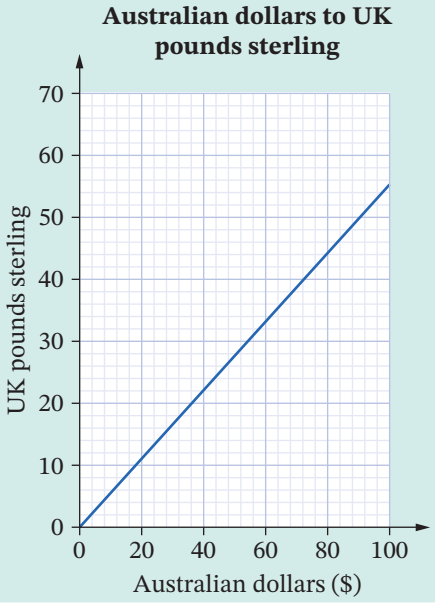
Example 5

Exchange rates change daily, but suppose the exchange rate between the Australian dollar and the UK pound sterling is \$A1 = £0.553, then \$A100 = £55.30 sterling. These values are used to draw this conversion graph.

Use the graph to convert:

a \$A50 to pounds

b £10 to Australian dollars.



Solution

Reading from the graph:

a \$A50 ≈ £28

b £10 ≈ \$A18

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EXERCISE 7.03 ANSWERS ON P. 635

Conversion graphs U F R C

EXAMPLE 5

1 Use the graph from the previous example for this question.

a Convert to pounds:

i $A40 **ii** $A88

b Convert to Australian dollars:

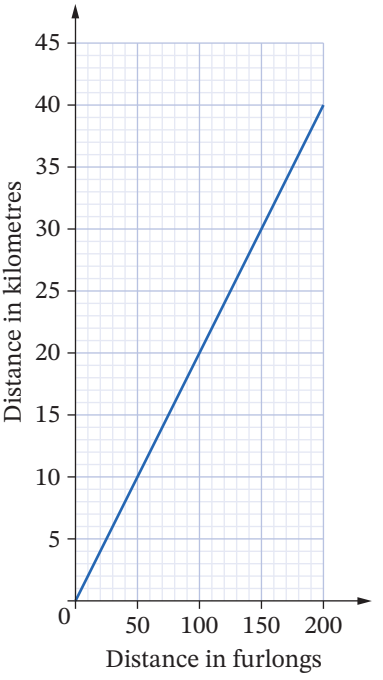
i £18 **ii** £60

c In 2015, $A1 = £0.50. R C

i How much less was $A40 worth in UK pounds sterling in 2015 than it is using this more recent conversion graph?

ii How much more was £60 worth in Australian dollars in 2015 for UK tourists visiting Australia, than it is using this more recent conversion graph?

2 The furlong is an imperial measure once used to measure length. This conversion graph shows distances in furlongs converted to kilometres.



a Convert to kilometres:

i 10 furlongs

ii 100 furlongs

iii 170 furlongs

b Convert to furlongs:

i 10 km

ii 25 km

iii 36 km

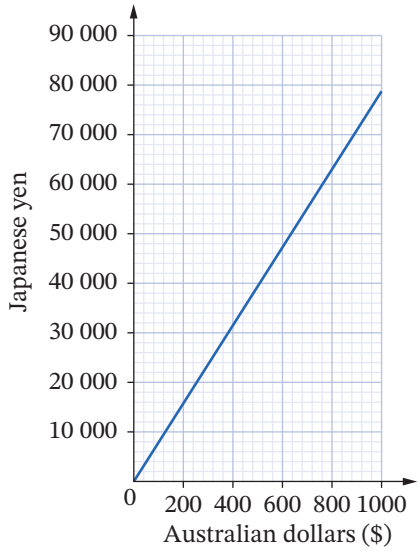
c Use an answer from part **a** to convert 300 furlongs to kilometres. R

d Use an answer from part **b** to convert 100 kilometres to furlongs. R

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3 This graph converts between Australian dollars and Japanese yen (¥).



a Convert to Japanese yen:

i $200

ii $800

iii $1000

b Convert to Australian dollars.

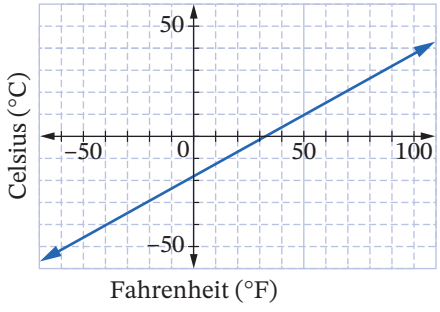
i ¥20 000

ii ¥60 000

iii ¥72 000

4 This graph converts temperatures between degrees Fahrenheit and degrees Celsius. Convert:

Degrees Fahrenheit to degrees Celsius



a 0°F to °C

b 50°F to °C

c 80°F to °C

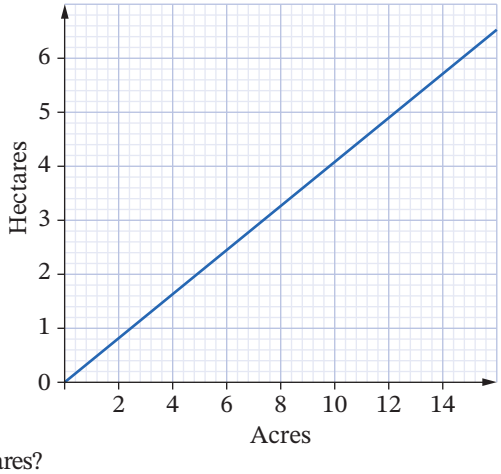
d 0°C to °F

e −10°C to °F

f 30°C to °F

5 This conversion graph converts between acres and hectares. The acre is an imperial measure of land area.

Converting acres to hectares



a Use the graph to convert 12 acres to hectares.

b A garden has an area of 5 acres. What is this area in hectares?

c Use the graph to convert 4.4 hectares to acres.

d Mr Ferguson has a property with an area of 5 hectares. How big is this in acres?

e A rectangular playing field measures 250 m by 128 m.

i What is the area of the field in square metres?

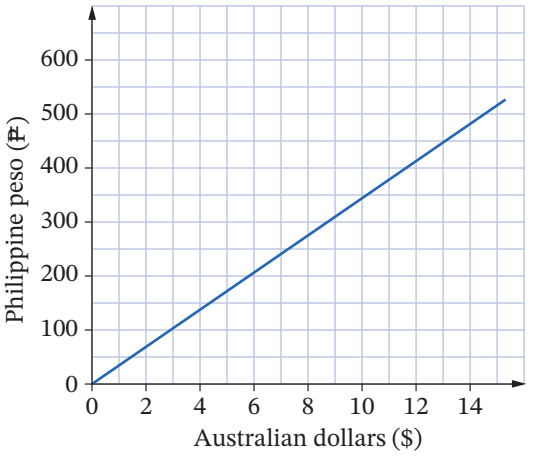
ii What is the area of the field in hectares?

iii What is the area of the field in acres?

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PATH

6 This graph is used to convert Australian dollars to Philippine pesos (₱).



a Change each Australian dollar amount into Philippine pesos.

i $15

ii $6

iii $9

b Change ₱350 to Australian dollars.

c How many Australian dollars would you receive for ₱200?

d Estimate the number of pesos you should get for $120. R

7.04 Distance–time graphs

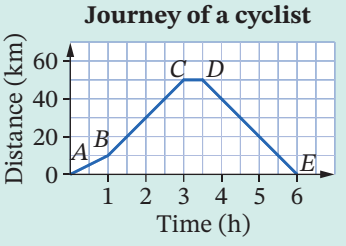
PATH

Puzzle
Jane's diary

Distance travelled by an object can be shown on a **distance–time graph** or **travel graph.**

Example 6

This distance–time graph shows the journey of a cyclist on a training ride.



a What was the total distance covered?

b The cyclist's speed increases at *B*, after the first hour. How is this shown by the graph?

c Calculate the speed of the cyclist from:

i *B* to *C* **ii** *D* to *E*

What do you notice?

d When was the cyclist stationary?

Solution

a The cyclist travelled 50 km, then returned to the starting point, so the total distance covered was 100 km.

b The graph becomes steeper after *B*.

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c Speed $= \frac{\text{distance travelled}}{\text{time taken}}$ or the gradient of the interval on the graph

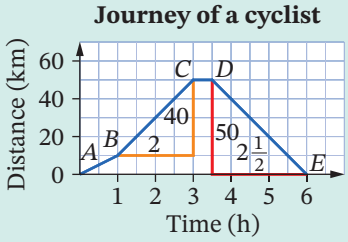
i Speed from B to $C = \frac{40}{2}$

$= 20$ km/h

ii Speed from D to $E = \frac{50}{2\frac{1}{2}}$

$= 20$ km/h

The speeds were the same.



d The cyclist was stationary (not moving) where the travel graph was flat, that is, between 3 and 4 hours (CD on the graph).

Distance–time graphs

- The gradient of the line shows the speed of the object.
- The steeper the graph, the greater the speed.
- If the gradient of the line is positive, the object is moving away from the starting point.
- If the gradient of the line is negative, the object is moving back towards the starting point.

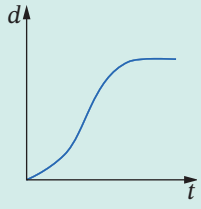
Distance–time graphs with variable speed

Example **6** showed a distance–time graph with straight lines, indicating that the speed of the object did not change much, staying fairly constant. However real-life situations are more complex and involve variable speed or a rate of change that is **not** constant. In these cases, the graph will be curved.

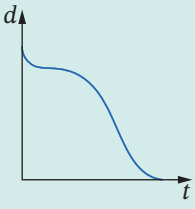
Example 7

Describe the journey represented by each distance–time graph.

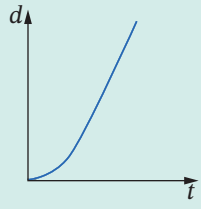
a



b

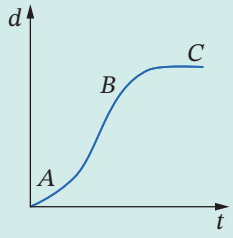


c



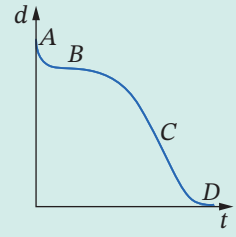
Solution

a The person starts the journey slowly (at A, the graph is not very steep), then increases his speed to a maximum (at B, the graph is the steepest), then slows down and stops (at C, graph becomes horizontal).

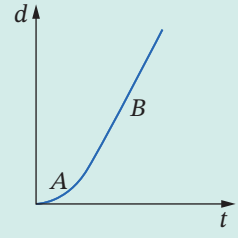


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b The person going home starts the journey at high speed (at *A*, the graph is very steep) before slowing down and stopping briefly (at *B*, the graph is flat). The person then speeds up again (at *C*, the graph becomes very steep) before slowing down again and arriving home (at *D*).



c The person starts the journey slowly, then speeds up (at *A*) before reaching a constant speed (at *B*, the graph is a straight line) and continuing on the journey.



Example 8

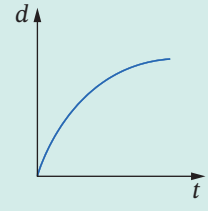
Draw distance–time graphs to represent each situation described.

a Kobi was running quickly but then his speed decreased until he stopped.

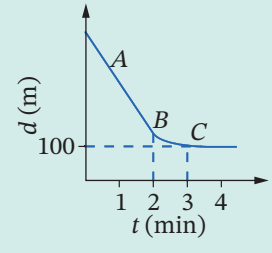
b Susanna was running home at a constant speed for 2 minutes, before slowing down and stopping 100 m from home after a further 1 minute.

Solution

a Kobi's speed is gradually decreasing, so the steepness (or gradient) of the graph must be steep at first, then decreasing.



b The graph should be a decreasing straight line (at *A*) for 2 minutes, indicating a constant speed. Then for one minute (at *BC*), the gradient of the graph should gradually decrease, until becoming flat (at *C*) to show when Susanna stops, 100 m from home.



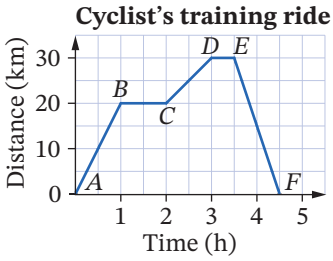
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EXERCISE 7.04 ANSWERS ON P. 635

Distance–time graphs U F R C

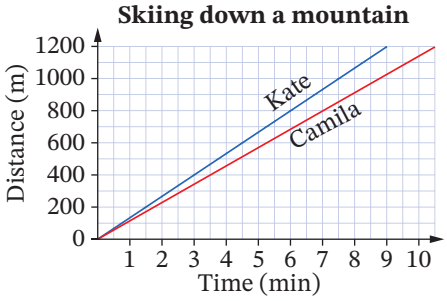
EXAMPLE 6

1 This graph shows a cyclist's journey on a training ride. R C



- **a** Describe the journey of the cyclist, giving the speeds at each stage.
- **b** Do any intervals of the graph indicate that the cyclist is travelling at the same speed? Give reasons.
- **c** The gradient of the interval *EF* is −30, but the speed at this stage is 30 km/h. What does the negative gradient indicate?

2 Kate and Camila are downhill skiers. Here is the distance–time graph for their 1200 m trip down the mountain. R C

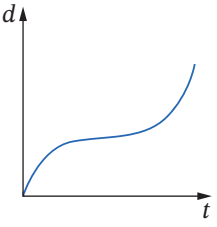


- **a** What was the speed of each skier?
- **b** Who reached the base of the mountain first and how many minutes did it take?
- **c** How many minutes later did the second skier arrive?
- **d** How far had Camila gone after 6 minutes?
- **e** How far *ahead* was Kate after 6 minutes?
- **f** If Camila and Kate were skiing *down* the mountain, why do the graphs go *up*?

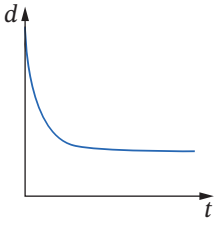
EXAMPLE 7

3 Describe the journey represented by each distance–time graph. R C

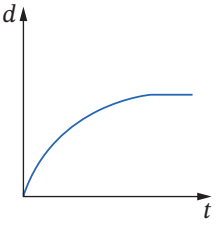
a



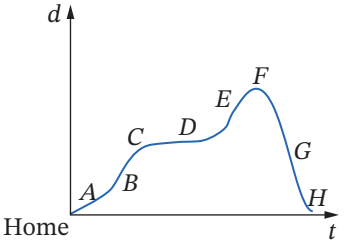
b



c



4 Match each statement to the correct part of the distance–time graph. R C



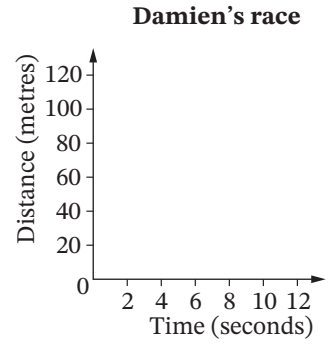
- **a** the person slows down because he is almost home
- **b** the person is not moving
- **c** after moving at a constant speed, the person increases his speed
- **d** the person changes direction for the return trip
- **e** after stopping, the person increases his speed
- **f** the person gradually slows down

PATH

EXAMPLE 8

5 Copy and complete this distance–time graph, using the following information about Damien completing a 100 m race. R C

- Damien covered 5 m at 2 seconds and 20 m at 4 seconds as his running speed increased
- Damien then ran at a constant speed until reaching the 80 m mark at 9 seconds
- Damien slowed down over the last 20 m, completing the 100 m race in 11 seconds
- Damien finally stopped after running a further 20 m



6 Sketch a distance–time graph for each situation described. R C

a Starting at home and running away from home at a decreasing speed

b Starting at home and running away from home at an increasing speed

c Starting away from home and running home at an increasing speed

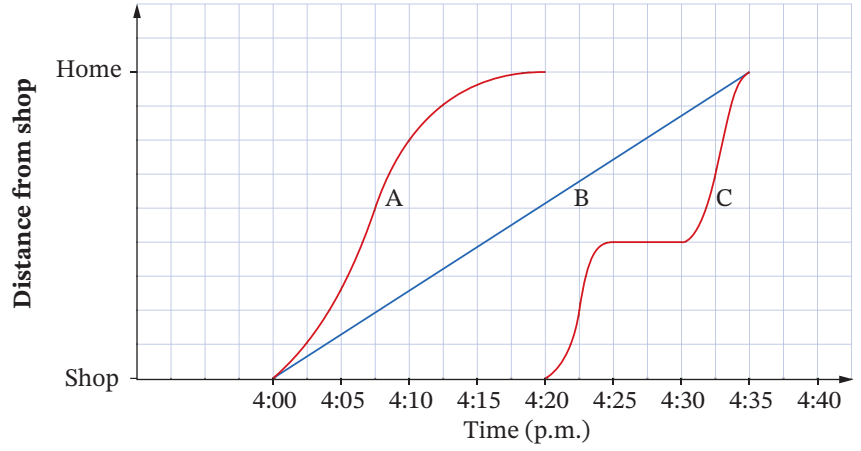
d Starting away from home and running home at a decreasing speed

7 These 3 statements can be matched to a graph below: R C

I Jade rode her bicycle home

II Cameron walked home

III Kiet ran home



a Match each statement to its correct graph.

b Explain in words why you made each match.

c What could have caused the level section at about 4:25 p.m. in graph **C**?

d Describe how the speed changes in graph **A**.

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8 Match each description to its correct graph. R C

a The speed increases at an increasing rate and then stops.

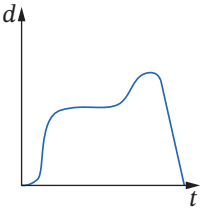
b The speed increases slowly, slows down to a stop, then increases to a constant rate.

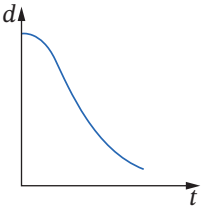
c The speed increases at an increasing rate, slows down and stops.

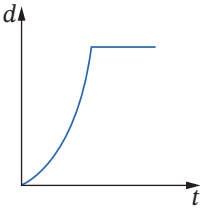
d The speed starts high, decreases, then stops.

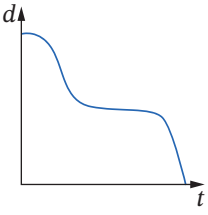
e The speed increases to a maximum, then slows down.

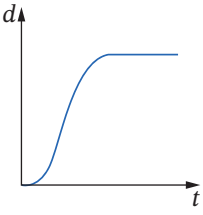
f The speed increases, then slows down and stops, then begins to increase again before stopping for an instant and returning to the start at a constant speed.

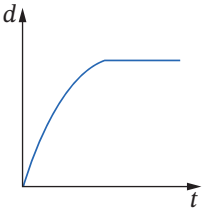
A


B


C


D


E


F


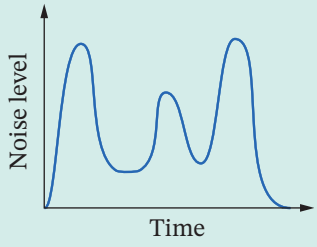
PATH

7.05

Graphs of variable change 7.05

Example 9

This graph shows the noise level of a classroom during a lesson. Describe what may have happened in the classroom during the lesson.



Solution

- The noise level increased as the students entered the classroom.
- The noise level lowered as the students settled down to work.
- There may have been a classroom discussion (the noise level increased) and then the class settled down again.
- The noise level increased towards the end of the lesson as students became restless.
- The period ended and the noise level decreased after the students left the classroom.

PATH

Worksheet
Graphs of change

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PATH

Video
Graphs of variable change

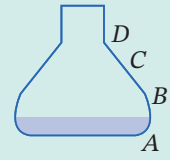
Example 10

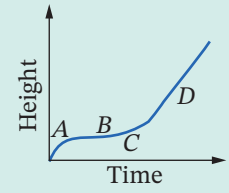
Water is poured into this container at a constant rate. Draw a graph to show the height of water rising in the container over time.



Solution

- The container is wider at B than at A, so the rate of increase in the water level will slow down.
- As the container narrows at C, the water level will increase faster (the graph becomes steeper).
- The container is cylindrical at D, so the water level will rise at a constant rate (the graph is a straight line).



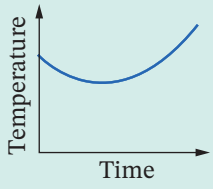


Video
Graphs of variable change

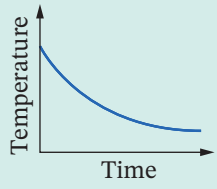
Example 11

A bowl of hot soup sits on the kitchen bench cooling. At first it loses heat quickly but, as time passes, it loses heat more slowly until it is at room temperature. Which of these graphs best illustrates this?

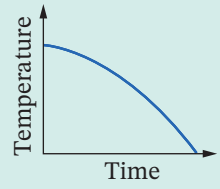
A



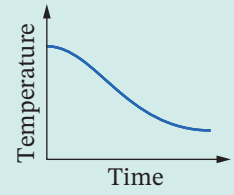
B



C



D



Solution

The correct graph is B. The temperature decreases rapidly initially (steep graph), but then this rate of decrease slows (graph becoming flatter, the temperature decreases at a decreasing rate).

A is incorrect, because it shows the temperature decreasing, then increasing.

C is incorrect because it shows the temperature decreasing slowly initially (flat graph), then decreasing quickly to 0 (steep graph, decreasing at an increasing rate).

D is incorrect because it shows the temperature decreasing slowly initially (flat graph), then quickly, then slowly again.

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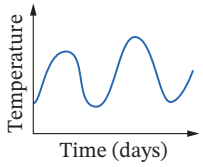
Graphs of variable change U F R C

EXAMPLE 9

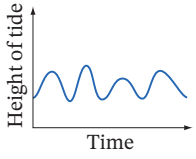
7.05

1 Describe what may be happening as represented in each graph. R C

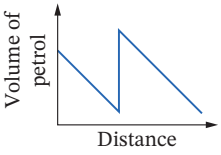
a



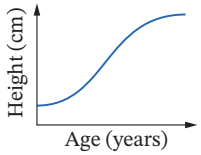
b



c



d

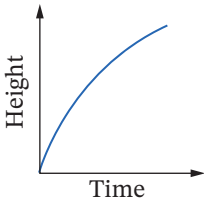


2 For each container, select the graph that best describes the height of the water as it is poured at a constant rate into the container. R

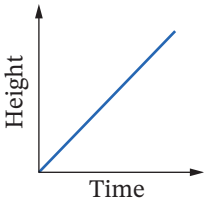
a



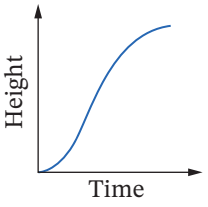
A



B



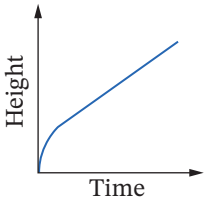
C



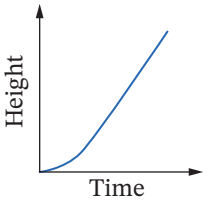
b



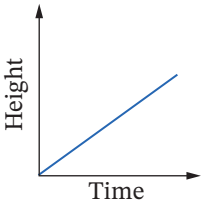
A



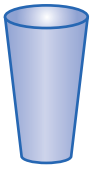
B



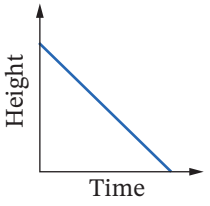
C



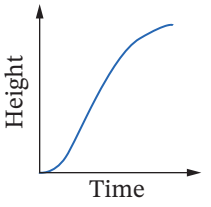
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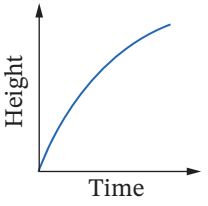
A



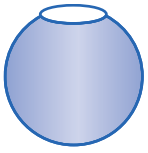
B



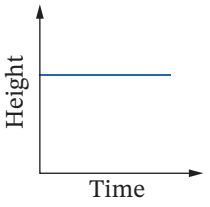
C



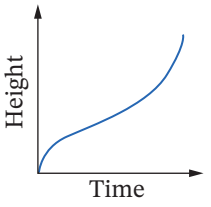
d



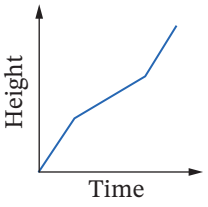
A



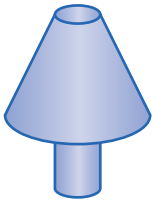
B



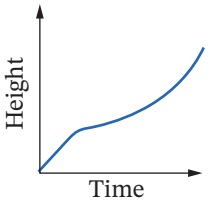
C



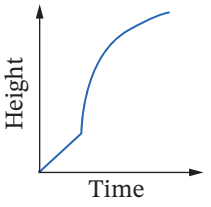
e



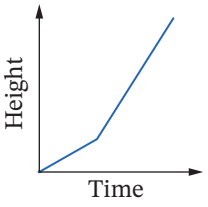
A



B



C



PATH

EXAMPLE 10

3 Draw a graph that models each situation described. Use the variables given in brackets. R C

a The water level in a pool with people swimming in it on a hot summer day. (height, time)

b The height of a girl above the ground as she climbs up the ladder of a slide, sits and then slides down. (height, time)

c The distance from a shop of a person walking at a steady speed up and down the street past that shop. (distance, time)

EXAMPLE 11

4 Match each story to its graph below. (The variables are given in brackets.) R

a People were purchasing from the drink-vending machine until it broke down. (number of cans in machine, time)

b The tank was half full of water all day. (depth of water, time)

c The cost of a mobile phone call is charged at a constant rate. (rate, time)

d Jo was watching a movie but stopped to take a phone call halfway through. (amount of movie seen, time)

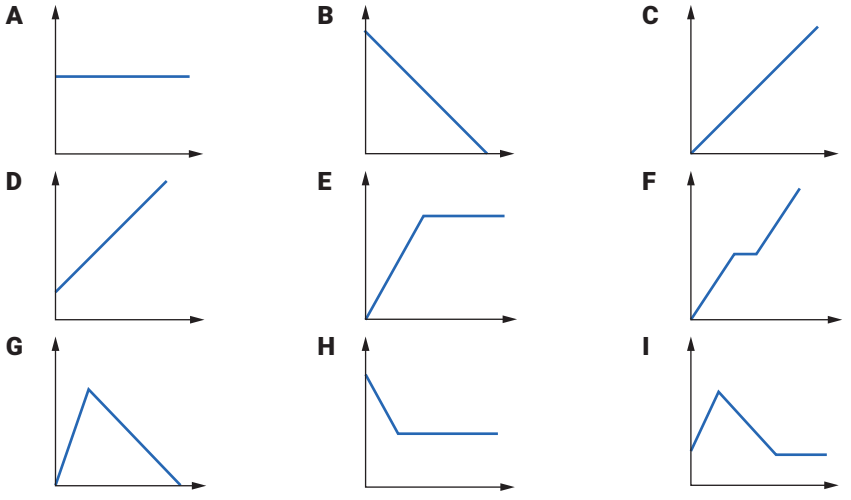
e Interest rates rise rapidly at a constant rate, then steadily fall at a constant rate and stay at a constant low. (rate, time)

f A taxi fare includes the hire charge plus a constant amount per kilometre. (cost, kilometres travelled)

g The car is consuming petrol at a steady rate until it runs out of fuel. (litres in tank, time)

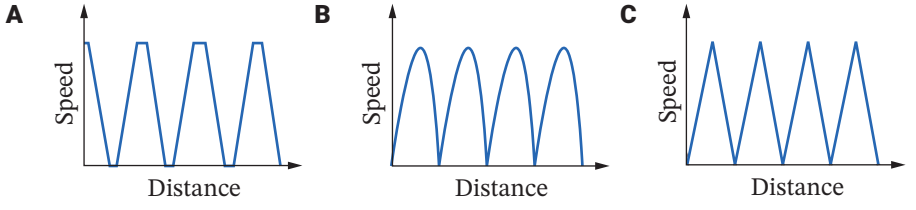
h Dario jogs at a steady rate, then stops and rests. (distance travelled, time)

i Jordan jogs at a steady rate, then walks home. (distance travelled, time)



5 For each description, select the graph that best describes the situation. Select the correct answer **A**, **B** or **C**. R

a The speed of a bus that stops 3 times

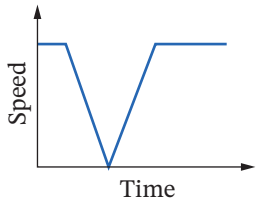


Foundation Standard Complex

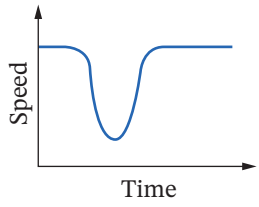
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b The speed of a car as it goes around a sharp corner

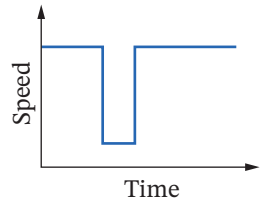
A



B

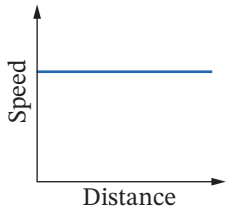


C

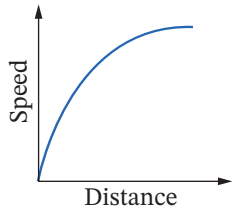


c The speed of a runner going up a steep hill from a standing start

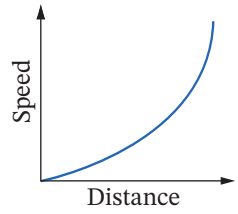
A



B

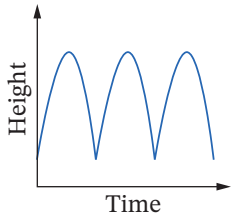


C

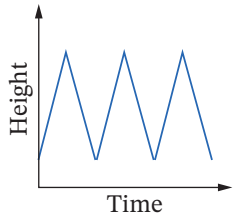


d The height of a person jumping on a trampoline

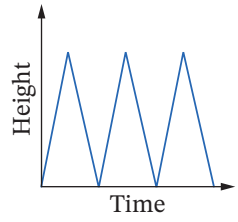
A



B



C



6 The 5 containers below are filled at a constant rate. Match a graph to each container, showing the water height against time. R

a



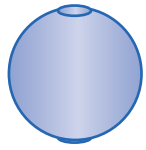
b



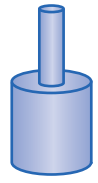
c



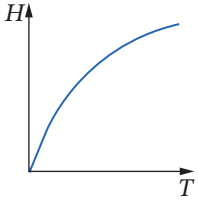
d



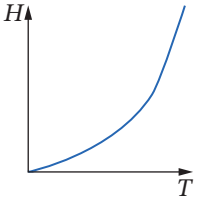
e



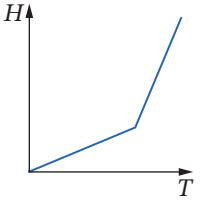
A



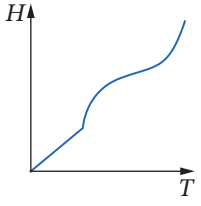
B



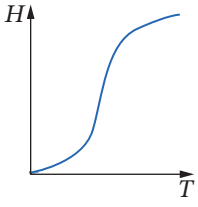
C



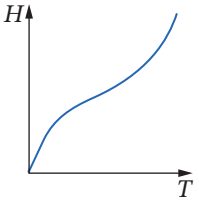
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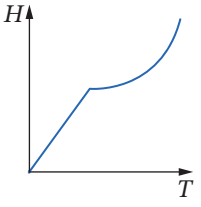
E



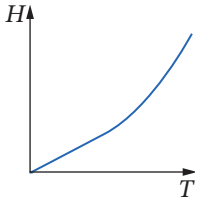
F



G



H



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PATH

7 Match each description to its correct temperature–time graph. R

a The temperature increases at a constant rate.

b The temperature decreases at a constant rate.

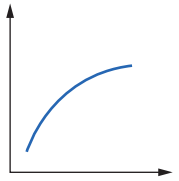
c The temperature increases at an increasing rate.

d The temperature decreases at a decreasing rate.

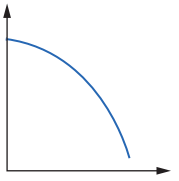
e The temperature increases at a decreasing rate.

f The temperature decreases at an increasing rate.

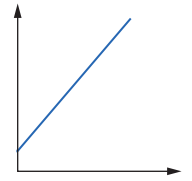
A



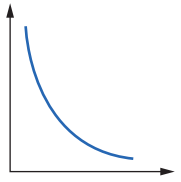
B



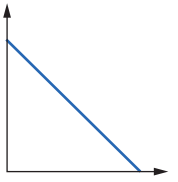
C



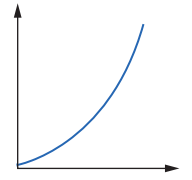
D



E



F



8 Select the speed–time graph that best describes: R

a a car accelerating until it reaches a constant speed

b a car stopped at traffic lights

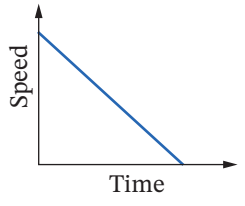
c a bus travelling at a constant speed before stopping for passengers

d a car travelling at a constant speed

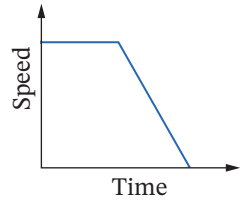
e a train slowing down at a constant rate until it stops

f a rocket launched into space

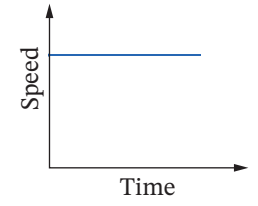
A



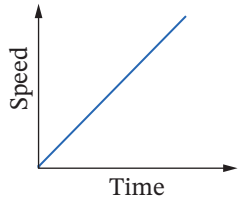
B



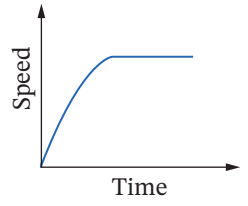
C



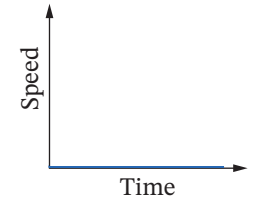
D



E



F



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Graphing quadratic equations $y = ax^2 + c$ 7.06

An equation in which the highest power of the variable is 2 is called a **quadratic equation**. The graph of a quadratic equation is a smooth U-shaped curve called a **parabola**.

Worksheet Graphing parabolas

Technology Investigating parabolas 1

Curve sketcher

Spreadsheets Investigating parabolas 1

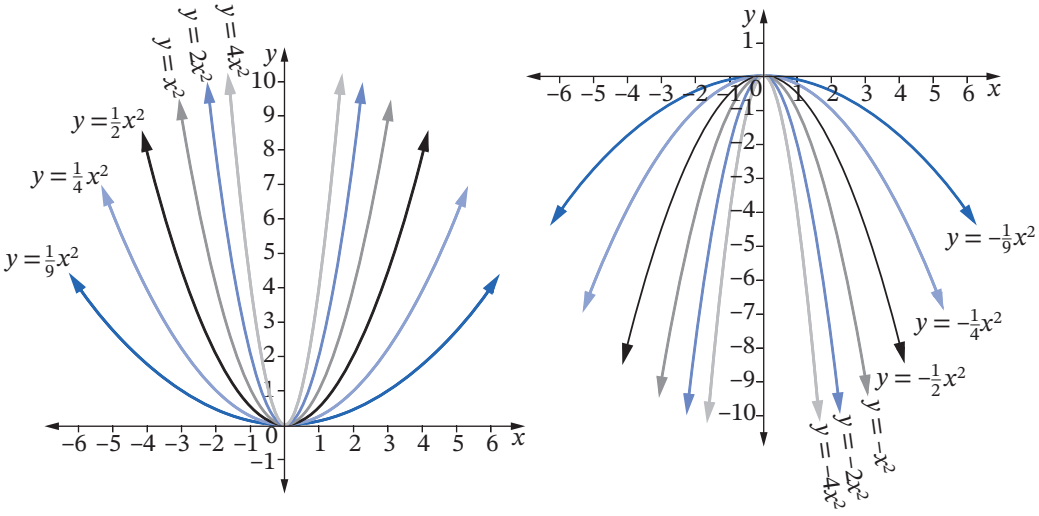
Curve sketcher

Interactive Graphing quadratics

The graph of $y = ax^2$

For the graph of $y = ax^2$, where a is a constant (number), the size of a (the **coefficient** of x^2) affects whether the parabola is 'wide' or 'narrow'.

As the size of a increases, the parabola becomes 'narrower' and as the size of a decreases, the parabola 'widens'. If a is negative, then the parabola is concave down.



$y = ax^2$ is a parabola with the y-axis as its **axis** (of symmetry) and (0, 0) as its **vertex**.

The graph of $y = ax^2 + c$

For the graph of $y = ax^2 + c$, where a and c are constants, the effect of c is to move the parabola up or down from the origin. Also, c is the y-intercept of the parabola.

Video
The parabola
$y = ax^2 + c$

Example 12

Graph each set of quadratic equations on the same number plane, showing the vertex of each parabola.

a $y = x^2, y = x^2 - 4, y = x^2 + 2$

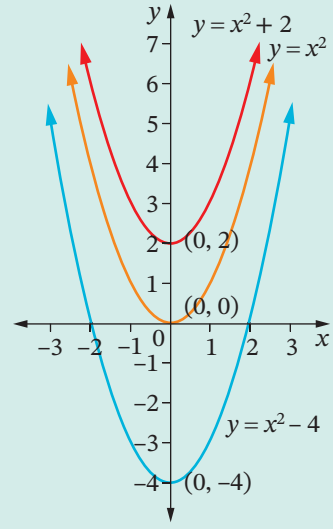
b $y = -x^2, y = -x^2 - 4, y = -x^2 + 5$

Solution

a First draw the graph of $y = x^2$. Its vertex is at (0, 0).

The graph of $y = x^2 - 4$ is like that of $y = x^2$, but moved 4 units down. Its vertex is at (0, –4).

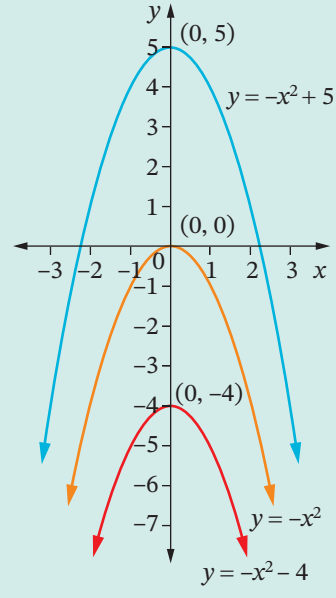
The graph of $y = x^2 + 2$ is like that of $y = x^2$, but moved 2 units up. Its vertex is at (0, 2).



b The graph of $y = -x^2$ is the graph of $y = x^2$ reflected across the x-axis. Its vertex is at (0, 0) as well.

The graph of $y = -x^2 - 4$ is like that of $y = -x^2$, but it is moved 4 units down. Its vertex is at (0, –4).

The graph of $y = -x^2 + 5$ is like that of $y = -x^2$, but it is moved 5 units up. Its vertex is at (0, 5).



In part **a**, all parabolas are concave up, because a is positive (1).

In part **b**, all parabolas are concave down, because a is negative (–1).

For $y = ax^2 + c$, the y-intercept of the parabola is c.

Example 13

For the graph of each quadratic equation, state:

- **i** whether the parabola is wider or narrower than the graph of $y = x^2$
- **ii** whether the parabola has moved up or down when compared to the graph of $y = x^2$
- **iii** the y-intercept.

a $y = 3x^2 - 1$ **b** $y = \frac{1}{3}x^2 + 2$

Solution

a i The coefficient of x^2 is 3, while the coefficient of x^2 in $y = x^2$ is 1.

$\therefore$ The parabola will be narrower than $y = x^2$.

ii The constant term is -1.

$\therefore$ The parabola has moved down.

iii The y-intercept is -1.

Vertex at $(0, -1)$

b i The coefficient of x^2 is $\frac{1}{3}$.

$\therefore$ The parabola will be wider than $y = x^2$.

ii The constant term is 2.

$\therefore$ The parabola has moved up.

iii The y-intercept is 2.

Vertex at $(0, 2)$

7.06

EXERCISE 7.06 ANSWERS ON P. 636

Graphing quadratic equations $y = ax^2 + c$ U F R C

This exercise can be completed using graphing technology.

EXAMPLE 12

1 a Graph each quadratic equation, showing the vertex of each parabola.

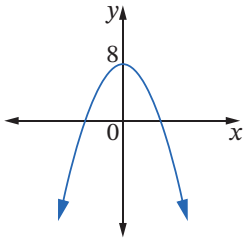
$y = x^2$ $y = -x^2$ $y = x^2 + 2$ $y = -2x^2$ $y = x^2 - 1$

b State which graphs you have drawn in part **a**: C

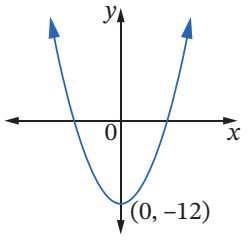
i are concave up **ii** are concave down **iii** have a turning point at $(0, 0)$.

2 Match each graph with its correct quadratic equation. R C

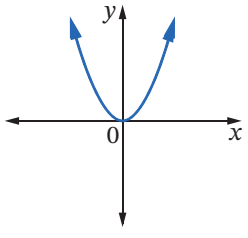
a



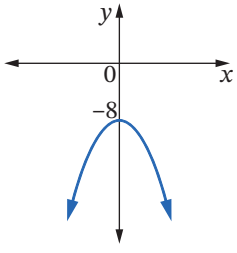
b



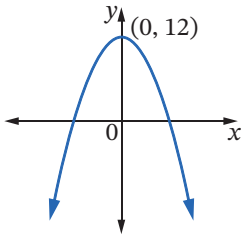
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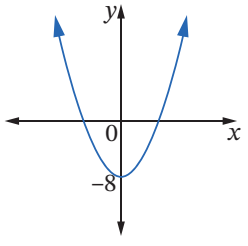
d



e

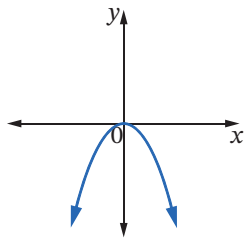


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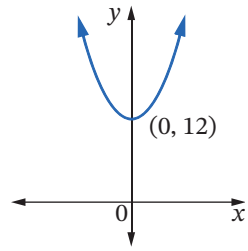


Foundation Standard Complex

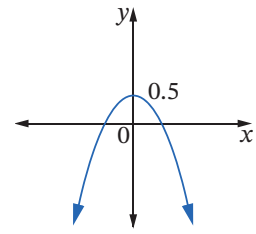
g



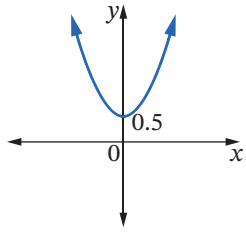
h



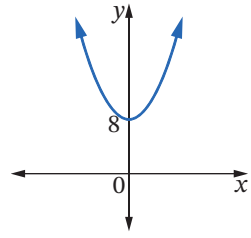
i



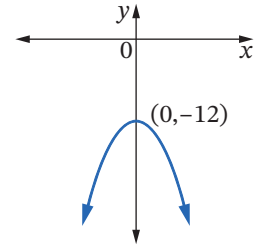
j



k



l



A $y = x^2$ **B** $y = -x^2$ **C** $y = x^2 - 8$ **D** $y = -12 - x^2$

E $y = \frac{1}{2} + x^2$ **F** $y = 8 - x^2$ **G** $y = 8 + x^2$ **H** $y = -x^2 + \frac{1}{2}$

I $y = x^2 - 12$ **J** $y = 12 - x^2$ **K** $y = -x^2 - 8$ **L** $y = x^2 + 12$

3 Find the equation of each parabola described, in the form $y = x^2 + c$ or $y = -x^2 + c$ (where c is a constant). R C

a vertex (0, 0), concave down

b concave up, turning point (0, 0)

c axis of symmetry $x = 0$, maximum $y = \frac{1}{4}$

d concave down, maximum $y = -9$

e turning point $\left(0, \frac{1}{2}\right)$, concave down

f axis of symmetry y-axis, minimum $y = 9$

4 a Graph $y = -3x^2 + 2$ after copying and completing this table. R C

x	−2	−1	0	1	2
y					

b Find the vertex.

c Write the equation of its axis of symmetry.

d Find its maximum value.

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5 Which statement is false about the graph of $y = 4x^2 - 1$? Select **A**, **B**, **C** or **D**. R C

A Its axis of symmetry is $y = 0$.

B It is concave up.

C The vertex is $(0, -1)$.

D It has a minimum value of $y = -1$.

6 For the graph of each given quadratic equation, state: R C

i whether the parabola is wider or narrower than the graph of $y = x^2$

ii whether the parabola has moved up or down when compared to the graph of $y = x^2$

iii the y-intercept.

a $y = 2x^2 + 3$

b $y = \frac{1}{2}x^2 + 1$

c $y = 6x^2 - 5$

d $y = 0.2x^2 - 12$

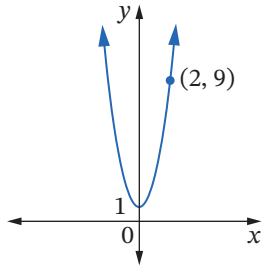
7 A parabola has the equation $y = x^2 - 5$.
Find the x-coordinates for the points on the parabola that have a y-coordinate of:

a 11

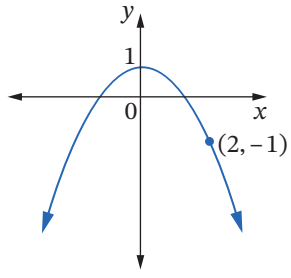
b 116.

8 Match each graph with its correct quadratic equation. R C

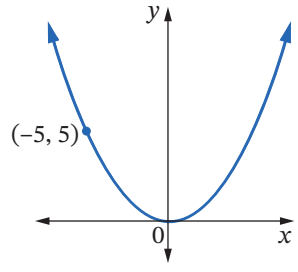
a



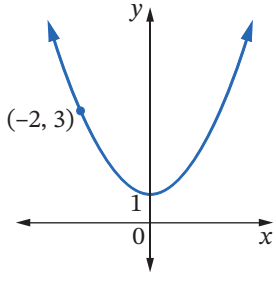
b



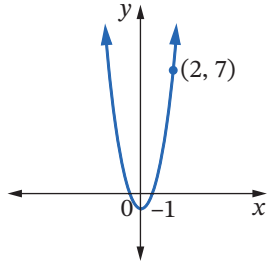
c



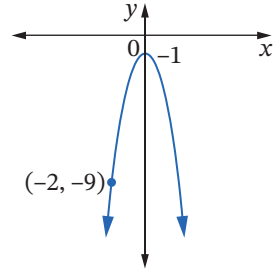
d



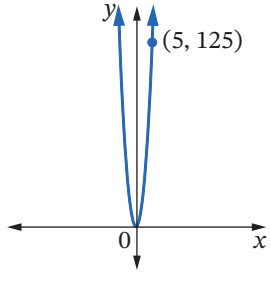
e



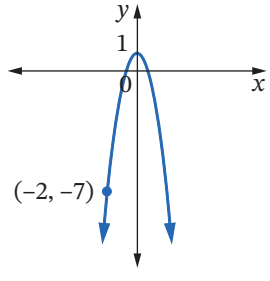
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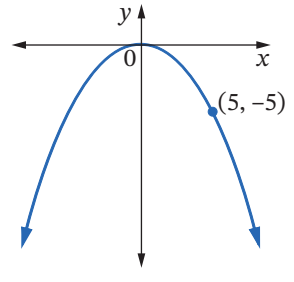
g



h



i

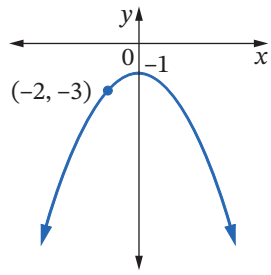


EXAMPLE 13

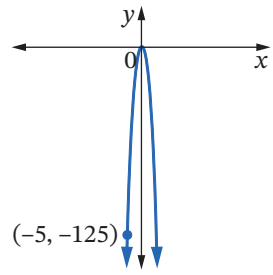
7.06

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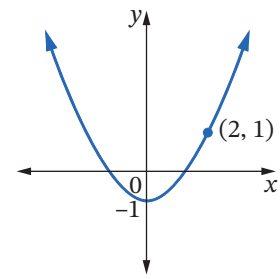
j



k



l



A $y = 5x^2$ **B** $y = 2x^2 + 1$ **C** $y = \frac{1}{2}x^2 - 1$

D $y = \frac{1}{5}x^2$ **E** $y = 2x^2 - 1$ **F** $y = -5x^2$

G $y = -\frac{1}{2}x^2 + 1$ **H** $y = -\frac{1}{5}x^2$ **I** $y = -2x^2 - 1$

J $y = \frac{1}{2}x^2 + 1$ **K** $y = -2x^2 + 1$ **L** $y = -\frac{1}{2}x^2 - 1$

Technology

Graphing $y = a(x - b)^2$

In this activity, use graphing technology to graph parabolas of the form $y = a(x - b)^2$.

1 a Use the software to graph the parabolas $y = (x - 1)^2$, $y = (x - 2)^2$, $y = (x + 2)^2$, $y = (x + 3)^2$.

b Compare and contrast the parabolas. What is the same and what is different?

c Are the parabolas concave up or down?

d Do they have a common vertex?

e What are their axes of symmetry?

f Given that these parabolas are of the form $y = (x + b)^2$, what does b represent?

2 a Use the software to graph the parabolas $y = 3(x - 2)^2$, $y = -(x - 4)^2$, $y = \frac{1}{2}(x + 3)^2$, $y = -0.1(x + 1)^2$.

b Compare and contrast the parabolas. What is the same and what is different?

c Which parabola is the widest? Which parabola is the narrowest?

d Which parabolas are concave up and concave down? Why?

e What is the vertex of each parabola?

f Given that these parabolas are of the form $y = a(x - b)^2$, what does b represent?

g What is the effect of the size of a on the shape of the parabola?

Graphing quadratic equations $y = a(x - b)^2 + c$

7.07

The parabola $y = a(x - b)^2 + c$

- The graph of $y = a(x - b)^2$ is a parabola with a vertex at $(b, 0)$.
- The parabola $y = a(x - b)^2$ is the graph of $y = ax^2$ translated b units to the right (or left if b is negative)
- The graph of $y = a(x - b)^2 + c$ is the parabola $y = a(x - b)^2$ translated c units up (or down if c is negative), so its vertex is at (b, c).

Example 14

Graph each parabola, clearly showing the vertex and y-intercept.

a $y = (x - 1)^2$ **b** $y = -(x + 2)^2$ **c** $y = 2(x + 1)^2 - 2$

Solution

a $y = (x - 1)^2$

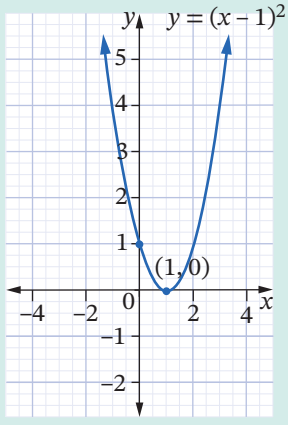
This parabola is the graph of $y = x^2$ translated 1 unit to the right, with a vertex at $(1, 0)$.

Substitute $x = 0$ for the y-intercept:

$y = (0 - 1)^2$

$= (-1)^2$

$= 1$



b $y = -(x + 2)^2$

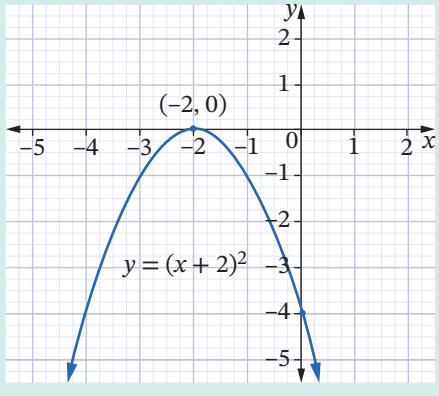
This parabola is the graph of $y = -x^2$ translated 2 units to the left, with a vertex at $(-2, 0)$.

It is concave down because a is negative.

Substituting $x = 0$ for the y-intercept:

$y = -(0 + 2)^2$

$= -4$



PATH

7.07

Worksheets
Matching parabolas

Matching parabolas with their equations

Technology
Investigating parabolas 2

Spreadsheet
Investigating parabolas 2

Video
Graphing the parabola $y = a(x - b)^2 + c$

PATH

c $y = 2(x + 1)^2 - 2$

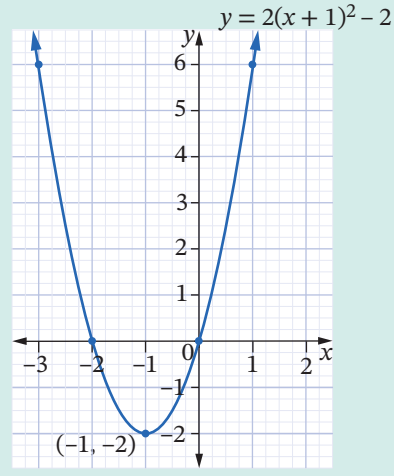
This is $y = 2x^2$ translated 1 unit to the left, with a vertex at $(-1, 0)$, then translated 2 units down, with a vertex at $(-1, -2)$.

Substituting $x = 0$ for the y-intercept:

$y = 2(0 + 1)^2 - 2$

$= 2 - 2$

$= 0$



EXERCISE 7.07 ANSWERS ON P. 637

Graphing quadratic equations $y = a(x - b)^2 + c$ U F R

This exercise can be completed using graphing technology.

EXAMPLE 14

1 Graph each parabola, clearly showing the vertex and y-intercept. R

a $y = (x - 3)^2$ **b** $y = (x - 2)^2$ **c** $y = (x + 1)^2 + 3$

d $y = -(x - 3)^2$ **e** $y = -(x + 1)^2$ **f** $y = -(x - 5)^2 - 1$

g $y = 3(x + 4)^2$ **h** $y = -2(x - 1)^2$ **i** $y = -\frac{1}{4}(x + 6)^2 + 2$

2 Which statement is true about the parabola $y = (x + 6)^2$?
Select the correct answer **A**, **B**, **C** or **D**. R

A Its axis of symmetry is $x = 6$ **B** It has a y-intercept at 36

C Its vertex is $(6, 0)$ **D** It passes through the point $(-1, 5)$

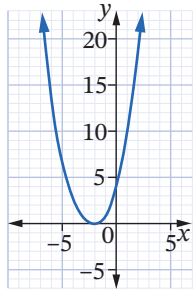
3 Match each equation to its graph below. R

a $y = (x + 4)^2$ **b** $y = -(x - 1)^2$ **c** $y = (x + 2)^2$

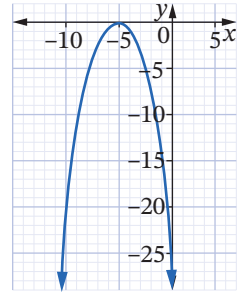
d $y = (x - 3)^2 + 4$ **e** $y = -(x + 5)^2$ **f** $y = -2(x + 1)^2$

g $y = -0.5(x - 4)^2$ **h** $y = 4(x - 1)^2 - 3$

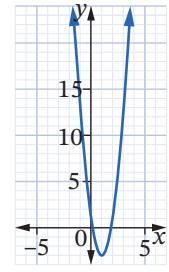
A



B



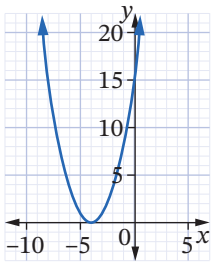
C



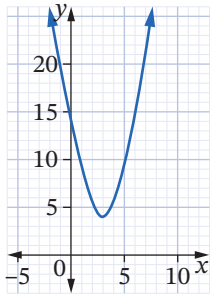
Foundation Standard Complex

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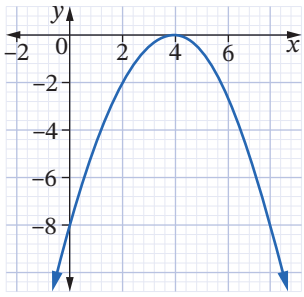
D



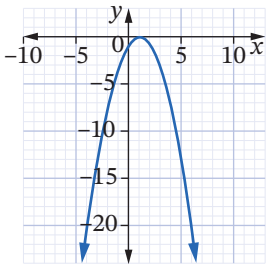
E



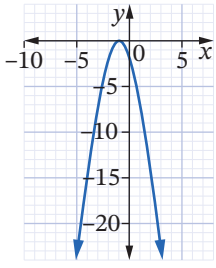
F



G



H



PATH

7.08

Applying quadratic equations 7.08

Parabolas can be seen and are used in our everyday lives. When water is sprayed upwards from a fountain or hose, or when a ball is thrown or kicked, the path of the object is a parabola.

Video
Applying quadratic functions

Example 15

A ball is thrown and its height (h metres) at any time (t seconds) is given by the quadratic equation $h = 2t - 0.5t^2$.

a Complete this table of values to graph the equation.

Time, t (seconds)	0	1	2	3	4	5
Height, h (metres)						

Quadratic equations can have different variables other than x and y.

b What is the maximum height of the ball?

c Find the total time the ball is in the air.

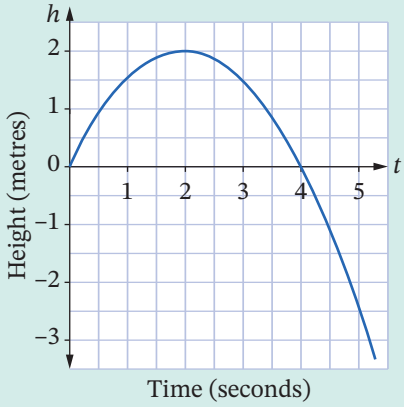
d Use the graph to find when the height of the ball is 1.5 m

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Solution

a

Time, t (seconds)	0	1	2	3	4	5
Height, h (metres)	0	1.5	2	1.5	0	−2.5



This is not the path of the ball. It is the graph of the height of the ball at different *times*, t seconds. We can ignore that part of the graph after $t = 4$ below the t-axis because the height cannot be negative.

b Maximum height is 2 m. From table or graph, $h = 2$

c 4 s When $h = 0$ again, at $t = 4$

d The ball is at a height of 1.5 m at 1 and 3 s.

EXERCISE 7.08 ANSWERS ON P. 637

Applying quadratic equations U F R C

EXAMPLE 15

1 A ball is thrown upwards and its height, h metres, after t seconds is given by the quadratic equation $h = 3.5t - 0.49t^2$. R C

a Copy and complete this table of values to graph the equation.

Time, t (seconds)	0	1	2	3	4	5	6	7	8
Height, h (metres)									

b What is the maximum height of the ball and when does it reach this height?

c At what times does the ball reach a height of 3 m? Explain why there are 2 answers.

2 The shape of cables hanging between 2 towers 100 m apart on a suspension bridge is modelled by the equation $h = 10 + 0.004x^2$ where h is the height, in metres above ground and x is the horizontal distance in metres from the centre of the bridge. R C

a Copy and complete this table, then graph the equation.

x	−50	−40	−30	−20	−10	0	10	20	30	40	50
h											

b How tall are the towers at either end of the cables?

c What is the height of the cables at their lowest point?

d How far from the centre is the height of the cables 15 m?

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3 A stone is dropped from a cliff and its height (h metres) at any time (t seconds) is given by $h = 80 - 4.9t^2$. R C

a Graph the quadratic equation for values of t from 0 to 5.

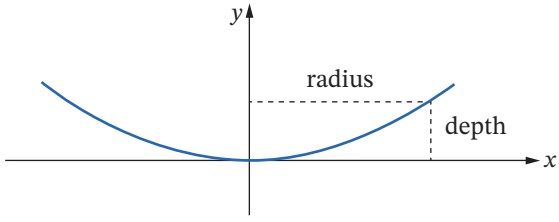
b What is the height of the cliff?

c Use your graph to find the height of the stone after 3 seconds.

d When will the stone hit the ground?

e How long after it is dropped is the stone 5 metres above the ground? Answer to the nearest 0.1 s.

4 The inside of a large radio telescope is the shape of a parabola and can be modelled by the equation $y = 0.0091x^2$, where y metres is the depth and x metres is the radius. R



a Copy and complete this table of values.

Radius, x (metres)	−40	−30	−20	−10	0	10	20	30	40
Depth, y (metres)									

b Graph the equation, using a scale of 1 cm = 10 m on the x-axis and 1 cm = 5 m on the y-axis.

c If the radius of the radio telescope is 32 m, use your graph to find its depth.

d Find the radius of the radio telescope that has a depth of:

i 1 m **ii** 2.5 m

5 A ball is dropped from the top of a building 200 m high and the distance, d metres, it falls after t seconds is given by the equation $d = 5t^2$. R

a Copy and complete this table of values and graph the equation.

Time, t (seconds)	0	1	2	3	4	5	6	7
Distance, d (metres)								

b How far has the ball dropped after:

i 2 seconds? **ii** 5 seconds?

c How far did the ball drop in:

i the 3rd second? **ii** the 4th second?

d After how many seconds did the ball hit the ground (correct to 1 decimal place)?

e After how many seconds was the ball 80 m above the ground?

6 A parabolic reflector has the equation $y = 0.2x^2$, where x (cm) is the radius of the reflector and y (cm) is the depth.

a Copy and complete this table and graph the equation.

Radius, x (cm)	−5	−4	−3	−2	−1	0	1	2	3	4	5
Depth, y (cm)											

b If the radius of a reflector is 2.5 cm, what is its depth?

c If the depth of a reflector is 4.5 cm, what is its radius?

7 At a water fountain, the height of the water being sprayed up follows the equation $h = 1.25 + 4t - 0.85t^2$, where h (metres) is the height of the water and t (seconds) is the time. R C



iStock.com/Petra Richli

a Copy and complete this table, then graph this equation.

Time, t (seconds)	0	1	2	3	4	5	6
Height, h (metres)							

b What is the vertical intercept of the graph and what does this value represent?

c How long does it take for the water to leave the fountain to when it hits the ground?

d What is the maximum height the water reaches?

e How long does it take for the water to reach its maximum height?

PATH

8 A ball is thrown upwards and its height (y metres) after x seconds is given by the equation $y = 8 - 0.2(x - 2)^2$. R C

a Copy and complete this table of values and then draw the graph of the equation.

Time (x seconds)	0	1	2	3	4	5	7	9
Height (y metres)								

b What is the y-intercept and what does it represent?

c What is the maximum height of the ball and when does it occur?

d When is the ball at a height of 6 metres?

e Find the height of the ball after

i 1.5 seconds **ii** 5 seconds

f How far does the ball travel in the

i 3rd second **ii** 6th second

g After how many seconds did the ball hit the ground?

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9 The path of a golf ball being hit off the tee is given by the quadratic equation $h = -0.004(x - 100)^2 + 40$, where h metres is the height of the ball and x metres is the horizontal distance the ball has travelled. R C

a Copy and complete the table of values below and then draw the graph of the equation.

Horizontal distance, x	0	20	40	60	80	....	200	220
Height, h								

b What is the y-intercept and what does it represent?

c What is the maximum height of the ball?

d How far from the tee does the ball hit the ground?

e At what distance does the golf ball reach a height of:

i 10 m? **ii** 30 m?

PATH

7.08

Quiz
Mental skills 7

Mental skills 7: Maths without calculators ANSWERS ON P. 640

Multiplying decimals

1 Study this example.

$3 \times 8 = 24$, so $3 \times 0.8 = 2.4$

0 dp + 1 dp = 1 dp (dp = decimal places)

The number of decimal places in the answer is equal to the total number of decimal places in the question. Also, the answer sounds reasonable because, by estimation:

$3 \times 0.8 \approx 3 \times 1 = 3$ $(2.4 \approx 3)$

2 Now evaluate each product.

a	0.7×5	**b**	12×0.2	**c**	0.4×0.3	**d**	$(0.6)^2$
e	8×0.1	**f**	0.03×0.9	**g**	4×0.05	**h**	1.1×8
i	0.3×0.8	**j**	0.2×0.06	**k**	9×0.2	**l**	0.07×0.4

3 Study this example.

Given that $15 \times 23 = 345$, evaluate 0.15×2300.

$0.15 \times 2300 = 0.15 \times 23 \times 100 = 3.45 \times 100 = 345$

2 dp + 0 dp = 2 dp

(*Estimate* $0.15 \times 2300 \approx 0.2 \times 2300 = \frac{1}{5} \times 2300 = 460$)

4 Now given that $39 \times 17 = 663$, evaluate each product.

a	3.9×17	**b**	39×170	**c**	39×0.17	**d**	0.39×1.7
e	3.9×1.7	**f**	390×1.7	**g**	3.9×0.17	**h**	3.9×170
i	3900×1.7	**j**	39×1.7	**k**	39×0.017	**l**	0.39×0.17

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Technology

Graphing $y = ax^3 + c$

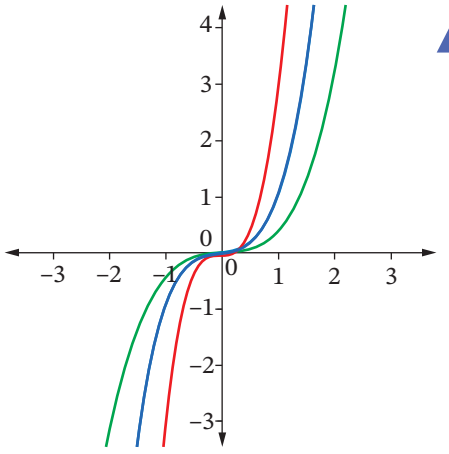
1 Use graphing technology to graph the cubic equations $y = 0.4x^3$, $y = x^3$ and $y = 3x^3$ in different colours.

2 What is the effect of a in $y = ax^3$?

3 Now graph $y = 2x^3$, $y = 2x^3 - 4$ and $y = 2x^3 + 1$ and compare the curves.

4 Graph $y = -\frac{1}{2}x^3$, $y = -\frac{1}{2}x^3 - 2$ and $y = -\frac{1}{2}x^3 + 3$ and compare the curves.

5 What is the effect of c in $y = ax^3 + c$?



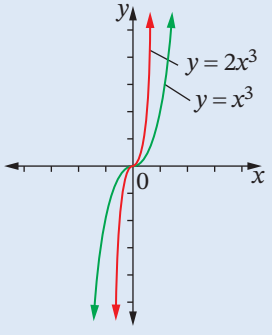
7.09 Graphing cubic equations $y = ax^3 + c$

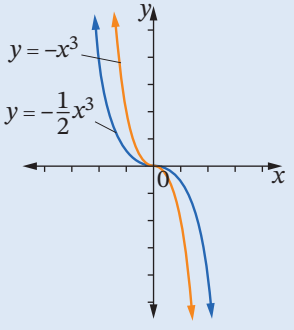
PATH

Worksheet
Graphing cubics 1

An equation in which the highest power of the variable is 3 is called a **cubic equation**, for example, $y = x^3 + 2x^2 - 5x + 10$. The graph of a cubic equation is called a **cubic curve**.

The cubic curve $y = ax^3$

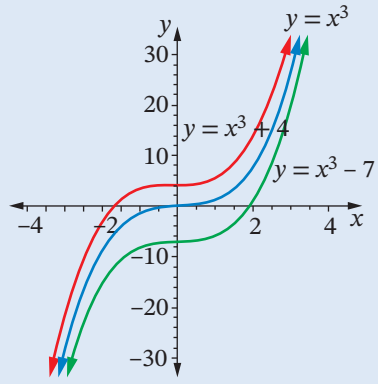




- The graph has no line symmetry.
- The graph has rotational symmetry of 180° about (0, 0): if you spin the curve upside-down, it maps onto itself.
- If a is positive, the graph is increasing, but flat at (0, 0)
- If a is negative, the graph is decreasing, but flat at (0, 0)
- The size of a (the coefficient of x^3) determines whether the cubic curve is 'wider' or 'narrower' (compared to $y = x^3$).

The cubic curve $y = ax^3 + c$

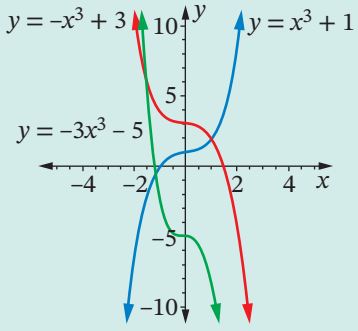
The graph of $y = ax^3 + c$ is the cubic curve $y = ax^3$ translated c units up (or down if c is negative), so its y-intercept is c).



Example 16

Graph the cubic equations $y = x^3 + 1$, $y = -x^3 + 3$ and $y = -3x^3 - 5$, showing the y-intercept of each curve.

Solution



EXERCISE 7.09 ANSWERS ON P. 640

Graphing cubic equations $y = ax^3 + c$ U F R C

1 Graph each cubic equation.

a $y = x^3 - 2$ **b** $y = -x^3$ **c** $y = 2x^3$

d $y = x^3 + 3$ **e** $y = -x^3 - 4$ **f** $y = -x^3 + 2$

g $y = 2x^3 - 3$ **h** $y = -3x^3 - 2$ **i** $y = \frac{1}{2}x^3 + 4$

EXAMPLE 16

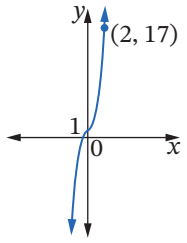
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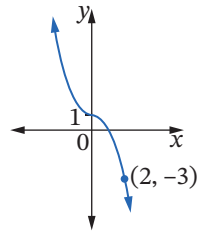
PATH

2 Match each cubic curve to its equation below. R

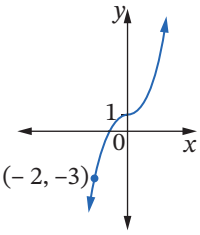
a



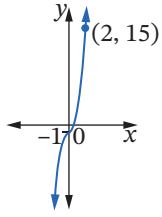
b



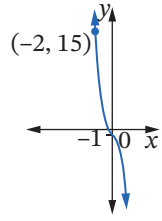
c



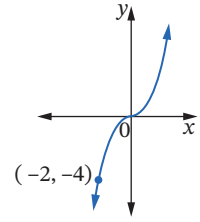
d



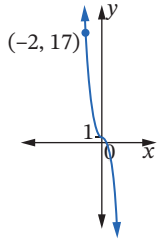
e



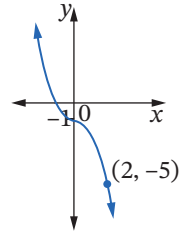
f



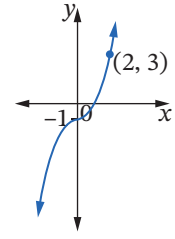
g



h



i



A $y = 2x^3 + 1$

B $y = \frac{1}{2}x^3 - 1$

C $y = 2x^3 - 1$

D $y = \frac{1}{2}x^3$

E $y = -\frac{1}{2}x^3 + 1$

F $y = -2x^3 - 1$

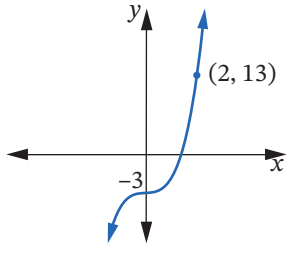
G $y = \frac{1}{2}x^3 + 1$

H $y = -2x^3 + 1$

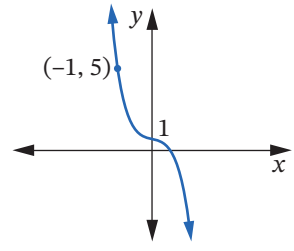
I $y = -\frac{1}{2}x^3 - 1$

3 Find the equation of each cubic curve in the form $y = ax^3 + c$. R C

a



b



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Technology

Graphing $y = x^n$

PATH

1 Use graphing technology to graph on the same axes $y = x^2$, $y = x^4$ and $y = x^6$.

2 How are the graphs similar? How are the graphs different?

3 Graph on the same axes $y = x^3$, $y = x^5$ and $y = x^7$.

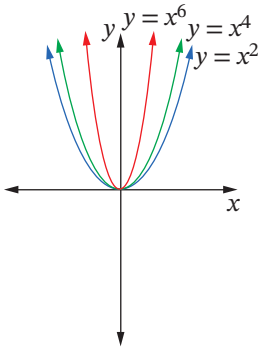
4 How are the graphs similar? How are the graphs different?

5 Describe the differences between the graphs of $y = x^n$ when n is even and when n is odd.

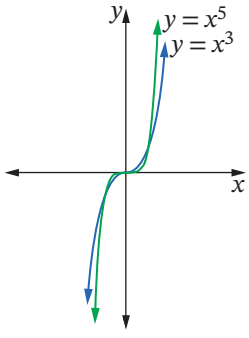
Graphing power equations $y = ax^n + c$ 7.10

We have already graphed $y = ax^2 + c$ and $y = ax^3 + c$. Now we will graph equations containing higher powers of x, of the form $y = ax^n + c$, where n is a positive integer.

The graphs of $y = x^2$, $y = x^4$ and $y = x^6$ are shown: even powers of x.



The graphs of $y = x^3$ and $y = x^5$ are shown: odd powers of x.



We can call these **power curves** as they are graphs of **power equations**.

PATH

Technology
Power functions

Spreadsheet
Power functions

The graph of $y = ax^n$ if n is even

- The even power curve is like a steeper parabola, symmetrical about the y-axis, with a vertex at (0, 0)
- The graph is concave up if a is positive and concave down if a is negative
- The higher the power (the value of n), the narrower the graph

The graph of $y = ax^n$ if n is odd

- The odd power curve is like a steeper cubic curve, with rotational symmetry of 180° about (0, 0)
- The graph has a steep gradient, except around (0, 0) where it is flat
- The higher the power (the value of n), the narrower the graph

PATH

The graph of $y = ax^n + c$

- The graph of $y = ax^n + c$ is the power curve $y = ax^n$ translated c units up (or down if c is negative), a vertical translation, so its y-intercept is c.

The graph of $y = a(x - b)^n$

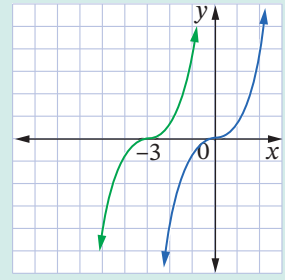
- The graph of $y = a(x - b)^n$ is the power curve $y = ax^n$ translated b units to the right (or left if b is negative), a horizontal translation.

Example 17

Sketch the graphs of $y = \frac{1}{2}x^3$ and $y = \frac{1}{2}(x+3)^3$ on the same axes.

Solution

The graph of $y = \frac{1}{2}(x+3)^3$ is obtained by moving the graph of $y = \frac{1}{2}x^3$ to the left by 3 units.



EXERCISE 7.10 ANSWERS ON P. 641

Graphing power equations $y = ax^n + c$ U F R C

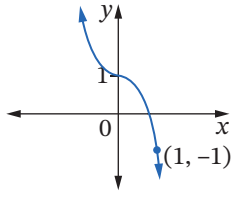
1 Match each equation to its correct graph. R

a $y = x^2 - 1$ **b** $y = (x-3)^3$ **c** $y = 2x^4 + 2$

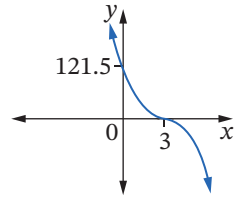
d $y = \frac{1}{3}(x+2)^3$ **e** $y = 2x^5 - 3$ **f** $y = -2x^3 + 1$

g $y = -(x-4)^6$ **h** $y = 3x^3 + 1$ **i** $y = -\frac{1}{2}(x-3)^5$

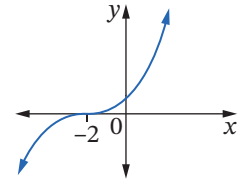
A



B



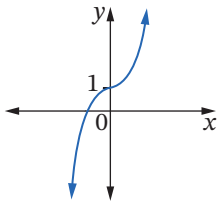
C



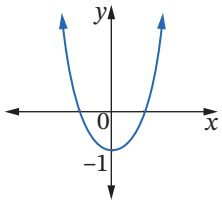
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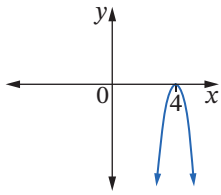
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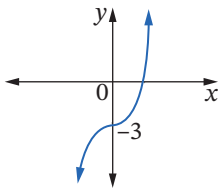
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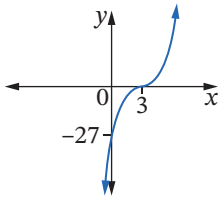
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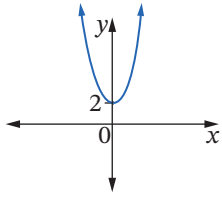
G



H



I



2 Sketch the graphs of each pair of equations. R

a $y = x^2$ and $y = (x - 2)^2$

b $y = 3x^2$ and $y = 3x^2 + 1$

c $y = -x^3$ and $y = -x^3 + 2$

d $y = -2x^4$ and $y = -2(x + 2)^4$

e $y = x^4$ and $y = (x - 1)^4$

f $y = -x^5$ and $y = -x^5 - 2$

EXAMPLE 17

3 For each pair of equations, explain how the second equation can be transformed using the graph of the first equation, for example, move left 4 units. R C

a $y = -x^5$ and $y = -x^5 + 4$

b $y = x^2$ and $y = (x - 5)^2$

c $y = 5x^6$ and $y = 5(x + 3)^6$

d $y = \frac{1}{4}x^3$ and $y = 4 + \frac{1}{4}x^3$

e $y = -x^4$ and $y = -(3 - x)^4$

f $y = -x^3$ and $y = -(x + 2)^3$

Investigation

Graphing $y = \frac{1}{x}$

1 Copy and complete this table for $y = \frac{1}{x}$. Explain why no y value exists for $x = 0$.

x	−5	−4	−3	−2	−1	−0.5	−0.2	−0.1	0	0.1	0.2	0.5	1	2	3	4	5
y									–								

2 Hence graph $y = \frac{1}{x}$ on a number plane, on paper or using graphing technology.

3 There are 2 parts or 'branches' to your graph. In which quadrants of the number plane are the branches?

4 Use your graph to explain what happens to the y value as x becomes very large.

5 Explain what happens to the y value as x approaches 0.

6 The graph of $y = \frac{1}{x}$ has 2 axes of symmetry. Draw them on your graph.

7 Copy and complete the table from question **1** for $y = -\frac{1}{x}$.

8 Hence graph $y = -\frac{1}{x}$ on a number plane.

9 How does the graph of $y = -\frac{1}{x}$ compare with that of $y = \frac{1}{x}$?

Foundation | Standard | Complex

7.11 Graphing hyperbolas $y = \frac{k}{x} + c$

PATH

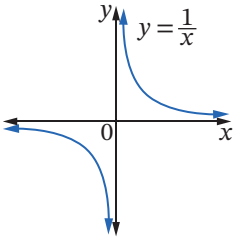
Worksheet
Graphing hyperbolas

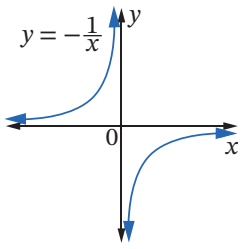
Technology
Curve sketcher

Spreadsheet
Curve sketcher

The graph of $y = \frac{k}{x}$, where k is a constant, is a curve with 2 branches called a **hyperbola** (pronounced 'hy-perb-o-la').

The graphs of $y = \frac{1}{x}$ and $y = -\frac{1}{x}$ are shown.





The hyperbola $y = \frac{k}{x}$

- The hyperbola has 2 separate branches in different quadrants.
- If k is positive, the graph is in the 1st and 3rd quadrants.
- If k is negative, the graph is in the 2nd and 4th quadrants.
- The graph has 2 axes of symmetry: their equations are the straight lines $y = x$ and $y = -x$.
- The graph has rotational symmetry of 180° about (0, 0).
- The higher the value of k, the further the hyperbola is from the x- and y-axes.
- As x becomes very large, y gets closer to 0.
- As x becomes closer to 0, y gets very large.
- The graph gets very close to the x- and y-axes but never crosses them. The x- and y-axes are called **asymptotes** because the graph approaches them but never touches them.

Video
Graphing hyperbolas

Example 18

Graph each hyperbola and mark the coordinates of one point on the curve.

a $y = \frac{2}{x}$ **b** $y = -\frac{3}{x}$

Solution

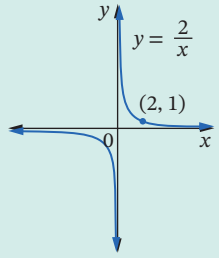
a Let $x = 2$

$y = \frac{2}{2}$

$= 1$

A point on the curve is (2, 1).

Choosing any value of x

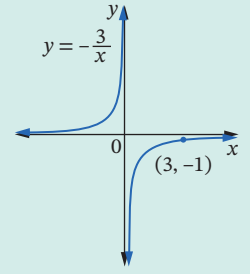


b Let $x = 3$

$y = \frac{3}{3}$

$= -1$

A point on the curve is (3, –1).



As $k = -3$ is negative, the hyperbola is in the 2nd and 4th quadrants.

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The hyperbola $y = \frac{k}{x} + c$

The graph of $y = \frac{k}{x} + c$ is the hyperbola $y = \frac{k}{x}$ translated c units up (or down if c is negative).

The hyperbola $y = \frac{k}{x-b}$

The graph of $y = \frac{k}{x-b}$ is the hyperbola $y = \frac{k}{x}$ translated b units to the right (or left if b is negative).

Example 19

Graph each hyperbola, find any intercepts and mark the coordinates of one point on the curve.

a $y = \frac{2}{x} + 1$ **b** $y = -\frac{3}{x-2}$

Solution

a Let $x = 2$

$y = \frac{2}{2} + 1$

$= 2$ $(2, 2)$.

The graph of $y = \frac{2}{x} + 1$ is the graph of $y = \frac{2}{x}$ translated up 1 unit. This means that the horizontal asymptote is now at $y = 1$.

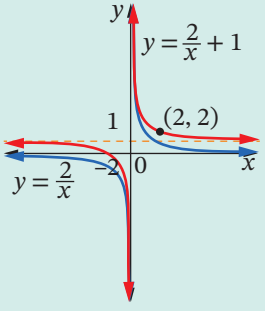
An x-intercept now occurs when $y = 0$.

$0 = \frac{2}{x} + 1$

$0 = 2 + x$ Multiplying both sides by x

$x = -2$

The x-intercept is -2.



PATH

b Let $x = 3$

$y = -\frac{3}{3-2}$

$= -3 \quad (3, -3).$

The graph of $y = -\frac{3}{x-2}$ is the graph of $y = -\frac{3}{x}$ translated right 2 units. This means that the vertical asymptote is now at $x = 2$.

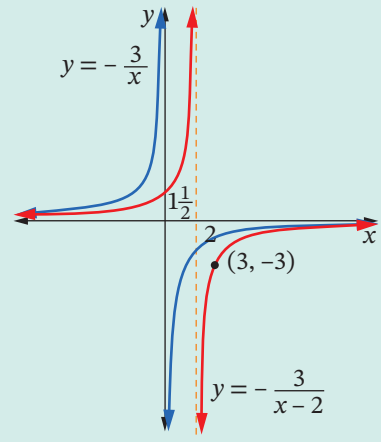
A y-intercept now occurs when $x = 0$.

$y = -\frac{3}{0-2}$

$= \frac{-3}{-2}$

$= 1\frac{1}{2}$

The y-intercept is $1\frac{1}{2}$.



EXERCISE 7.11 ANSWERS ON P. 641

Graphing hyperbolas $y = \frac{k}{x} + c$ UFRC

1 **a** Copy and complete this table for $y = \frac{2}{x}$.

x	−3	−2	−1	0	1	2	3
y				–			

b Graph $y = \frac{2}{x}$, showing the coordinates of one point on the hyperbola.

c On your diagram, draw in the axes of symmetry for the hyperbola.

d What are the equations of these axes?

EXAMPLE 18

2 Graph each hyperbola and mark the coordinates of one point on the curve.

a $y = \frac{4}{x}$ **b** $y = -\frac{2}{x}$ **c** $y = \frac{3}{x}$ **d** $y = -\frac{6}{x}$

3 **a** The distance from Sydney to Melbourne is close to 1000 km. Using the equation $s = \frac{1000}{t}$, copy and complete the following table that relates time (t hours) and speed (s km/h) for the trip. Round your answers to the nearest km/h.

t	1	2	3	...	10
s					

b Hence graph the equation $s = \frac{1000}{t}$.

c Why are the values for t only positive numbers? Explain why t cannot be equal to 0. R C

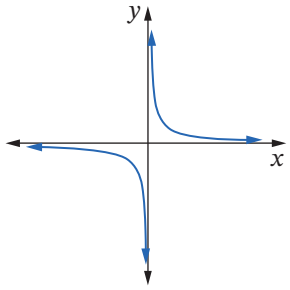
d If the time is doubled, is the speed halved? Use the information from your graph to support your answer. R C

Foundation Standard Complex

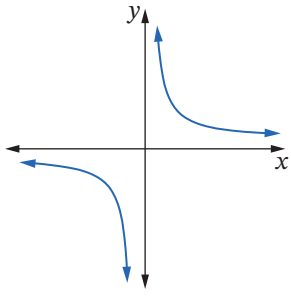
4 Match each equation with its graph. R

a $y = \frac{5}{x}$ **b** $y = -\frac{2}{x}$ **c** $y = \frac{1}{x}$ **d** $y = -\frac{5}{x}$

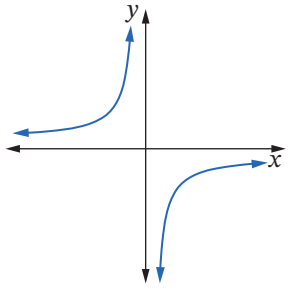
A



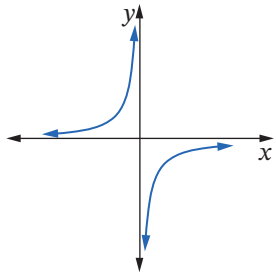
B



C



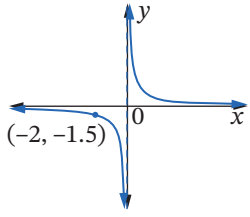
D



5 This curve is a hyperbola of the form $y = \frac{k}{x}$.

a Find the value of k.

b Hence state the equation of this hyperbola.



6 Graph each hyperbola and mark the coordinates of one point on the curve.

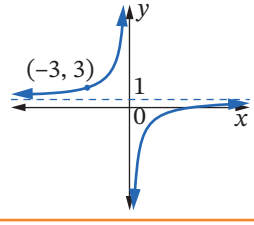
EXAMPLE 19

a $y = \frac{1}{x} + 2$ **b** $y = -\frac{2}{x} - 3$ **c** $y = \frac{2}{x-1}$ **d** $y = \frac{-3}{x+2}$

7 This curve is a hyperbola with equation $y = \frac{k}{x} + c$.

a Find the values of c and k.

b Hence state the equation of this hyperbola.



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8 Sarah and David want to buy a rectangular block of land that has an area of 800 m^2. There are several blocks available with this area. R C

a Copy and complete this table that relates the length (L metres) and width (W metres) of the block of land.

L	10	20	30	...	100
W					

b What is the formula for W?

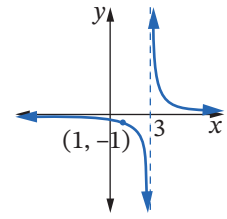
c Explain why the length or width cannot be equal to 0 metres.

d Graph the formula for W.

e What happens to the width as the length continues to increase? How is this shown on the graph?

f What happens to the width as the length approaches 0? How is this shown on the graph?

9 Which equation best represents this hyperbola?
Select the correct answer **A**, **B**, **C** or **D**.



A $y = \frac{2}{x-3}$ **B** $y = \frac{-1}{x+3}$

C $y = \frac{2}{x} + 3$ **D** $y = -\frac{1}{x} - 3$

10 Find the asymptotes of the hyperbola with equation: R C

a $y = \frac{k}{x}$ **b** $y = \frac{k}{x} + c$ **c** $y = \frac{k}{x-b}$ **d** $y = \frac{k}{x-b} + c$

Investigation

Graphing $y = 2^x$

This activity can be completed using graphing technology.

1 Copy and complete this table of values for $y = 2^x$.

x	−3	−2	−1	0	1	2	3	4
y								

2 Graph the points from the table and join them with a smooth curve. The equation $y = 2^x$ is called an exponential equation and its graph is called an exponential curve (exponent means 'power').

3 Graph $y = 2^{-x}$ in a similar way.

4 Compare the graphs of $y = 2^x$ and $y = 2^{-x}$. Describe any similarities and differences.

5 The y-intercept of any graph with equation $y = a^x$ (where a is a positive constant) is always 1. Explain why.

6 The graph of $y = 2^x$ is increasing. Is the graph of $y = 2^{-x}$ increasing or decreasing? Give reasons.

7 Describe what happens to the graph of $y = 2^x$ when:

a x approaches a large positive number

b x approaches a large negative number.

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Investigation

Graphing $y = 3(2^x)$

This activity can be completed using graphing technology.

1 Copy and complete this table of values for $y = 3(2^x)$.

$3(2^x)$ means $3 \times (2^x)$

x	−3	−2	−1	0	1	2	3	4
y								

2 Graph the points from the table and join them with a smooth curve

3 Graph each exponential equation in a similar way.

a $y = 1.5(2^x)$ **b** $y = 4(3^x)$ **c** $y = 0.5(4^x)$

4 Compare the graphs $y = 3(2^x)$, $y = 1.5(2^x)$, $y = 4(3^x)$ and $y = 0.5(4^x)$. Describe any similarities and differences.

5 Graph each exponential equation and describe any similarities and differences.

a $y = 5(2^{-x})$ **b** $y = 4(3^{-x})$ **c** $y = 0.5(4^{-x})$ **d** $y = 2(5^{-x})$

6 **a** What are the y-intercepts of the graphs in Question 1?

b What are the y-intercepts of the graphs in Question 2?

c What is the y-intercept of any graph with equation $y = k(a^x)$ or $y = k(a^{-x})$?

7.12

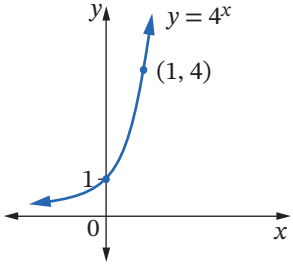
Graphing exponential equations $y = ka^x + c$

7.12

An equation of the form $y = a^x$, where a is a positive constant and the variable x is a power, is called an **exponential equation**, for example, $y = 5^x$, $y = 2^x$ and $y = 3^x$. The graph of an exponential equation is called an **exponential curve**.

The table of values and graph of $y = 4^x$ is shown.

x	−2	−1	0	1	2	3	4
y	$\frac{1}{16}$	$\frac{1}{4}$	1	4	16	64	256



- The y-intercept of $y = a^x$ is always 1 since $a^0 = 1$.
- As x increases (to the right, in the positive direction), a^x becomes very large. Graphically, this means that the graph of $y = a^x$ increases sharply with a steep positive gradient.
- As x decreases (to the left, in the negative direction), a^x approaches 0. This means that the graph of $y = a^x$ flattens out and approaches the x-axis as x becomes a large negative number. The x-axis is an **asymptote** because the curve approaches it but never touches it.
- The exponential curve is always above the x-axis because the value of a^x is always positive.

Videos
The exponential curve

The emperor's chessboard

Worksheet
Graphing exponentials

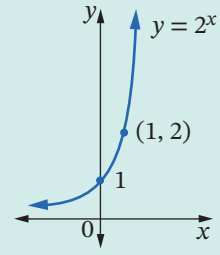
Example 20

Sketch each exponential equation and mark the y-intercept on each curve.

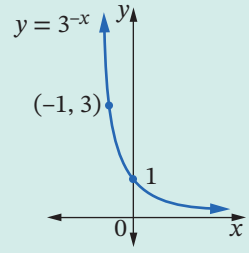
a $y = 2^x$ **b** $y = 3^{-x}$

Solution

a
- The y-intercept of $y = 2^x$ is 1
- When $x = 1$, $y = 2$, so (1, 2) is a point on the curve
- As x increases (to the right in the positive direction), 2^x becomes very large (steep positive gradient)
- As x decreases (to the left in the negative direction), 2^x approaches 0.
- The x-axis is an asymptote.



b
- The y-intercept of $y = 3^{-x}$ is 1
- When $x = -1$, $y = 3$, so (−1, 3) is a point on the curve
- As x decreases (to the left in the negative direction), 3^{-x} becomes very large (steep negative gradient)
- As x increases (to the right in the positive direction), 3^{-x} approaches 0.
- The x-axis is an asymptote.



Note that the graph of $y = 3^{-x}$ (and of $y = a^{-x}$ in general) is decreasing, and is actually a reflection of $y = 3^x$ in the y-axis.

PATH

Example 21

Sketch each exponential equation and mark the y-intercept on each curve.

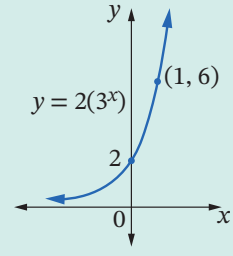
a $y = 2(3^x)$ **b** $y = 5(2^{-x})$

Solution

a The y-intercept of $y = 2(3^x)$ is 2.

(1, 6) is a point on the curve.

As x increases (to the right in the positive direction), $2(3^x)$ becomes very large.

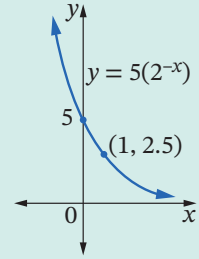


b The y-intercept of $y = 5(2^{-x})$ is 5.

(1, 2.5) is a point on the curve.

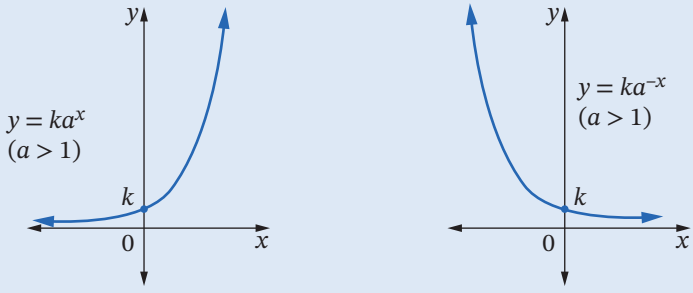
As x increases (to the right in the positive direction), $5(2^{-x})$ approaches 0.

Note that the y-intercept is the number to the left of a^x in the exponential equation.



The exponential curve $y = ka^x$ (where $a > 1$)

- The graph is always increasing.
- The y-intercept is k since $y = k(a^0) = k \times 1 = k$.
- The x-axis is an asymptote.
- As x increases (to the right), ka^x becomes very large, increasing sharply with a steep positive gradient.
- As x decreases (to the left), ka^x becomes very small, approaching 0 and the x-axis and flattening out.
- For $y = ka^{-x}$, the shape of the graph is reversed horizontally, or reflected in the y-axis, and the graph is always decreasing.



The exponential curve $y = ka^x + c$ (where $a > 1$)

- The graph of $y = ka^x + c$ is the exponential curve $y = ka^x$ translated c units up (or down if c is negative).
- The y-intercept is $k + c$ since $y = k(a^0) + c = k(1) + c = k + c$.
- The horizontal line $y = c$ is an asymptote.

Example 22

Graph each exponential equation, showing the y-intercept and the asymptote.

a $y = 2(3^x) + 5$ **b** $y = 2(3^x) - 5$ **c** $y = 2(3^{-x}) + 5$

Solution

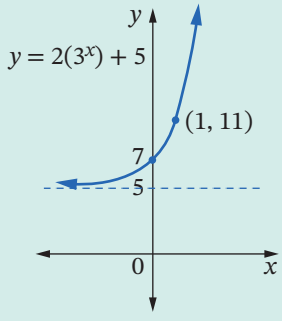
a Complete a table of values for $y = 2(3^x) + 5$

x	−3	−2	−1	0	1	2	3
y	5.07	5.22	5.67	7	11	23	59

The y-intercept of $y = 2(3^x) + 5$ is $2(3^0) + 5 = 2 + 5 = 7$.

(1, 11) is a point on the curve.

$y = 5$ is an asymptote.



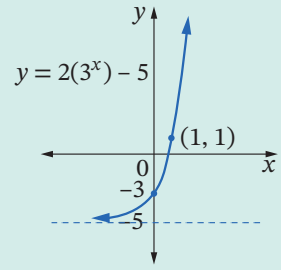
PATH

b Same shape as the graph in part **a** above.

The y-intercept is $y = 2(3^0) - 5 = 2 - 5 = -3$.

$(1, 1)$ is a point on the curve.

$y = -5$ is an asymptote.



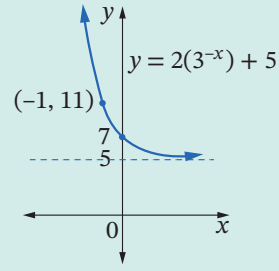
c $y = 2(3^{-x}) + 5$

x	−3	−2	−1	0	1	2	3
y	59	23	11	7	5.67	5.22	5.07

The y-intercept is $2(3^0) + 5 = 2 + 5 = 7$.

$(-1, 11)$ is a point on the curve.

$y = 5$ is an asymptote.



EXERCISE 7.12 ANSWERS ON P. 642

Graphing exponential equations $y = ka^x + c$ U F R C

Some of this exercise may be completed using graphing technology.

EXAMPLE 20

1 **a** Graph each exponential equation on the same axes. R C

i $y = 2^x$ **ii** $y = 3^x$ **iii** $y = 5^x$

b What is the y-intercept of each curve?

c Describe what happens to the graph of $y = a^x$ as a increases.

2 **a** Graph $y = 4^x$ and $y = 4^{-x}$ on the same axes.

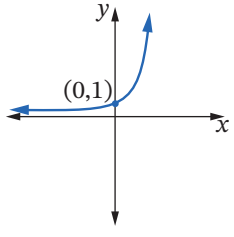
b Copy and complete: C

i The reflection of $y = 4^x$ in the y-axis is ...

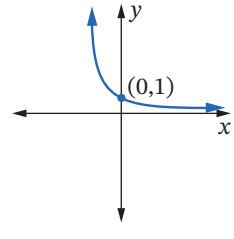
ii The reflection of $y = a^x$ in the y-axis is ...

3 Which graph represents $y = 2^{-x}$? Select the correct answer **A**, **B**, **C** or **D**. R C

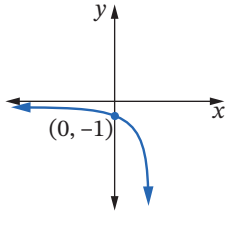
A



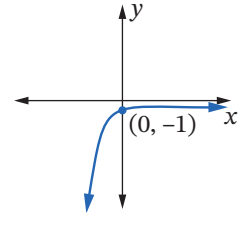
B



C



D



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4 **a** Graph $y = 2^x$ and $y = -2^x$ on the same axes.

b How are the 2 graphs related? R C

c Copy and complete: The reflection of $y = a^x$ in the x-axis is ...

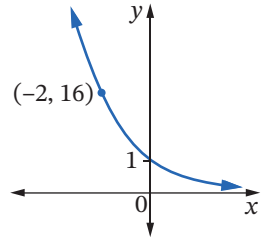
5 Graph $y = 3^x + 1$ and $y = 3^x - 1$ on the same axes and describe how they are related. R C

6 Sketch each exponential curve, showing the y-intercept.

a $y = 2^x$ **b** $y = 3^{-x}$ **c** $y = -4^x$

d $y = -2^{-x}$ **e** $y = 4^x + 1$ **f** $y = 4^x - 1$

7 Find an exponential equation for this graph. R C



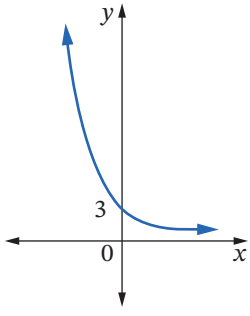
8 **a** Sketch each exponential curve, showing the y-intercept. R C

i $y = 4(3^x)$ **ii** $y = 2(5^x)$ **iii** $y = 3(2^{-x})$ **iv** $y = 5(4^{-x})$

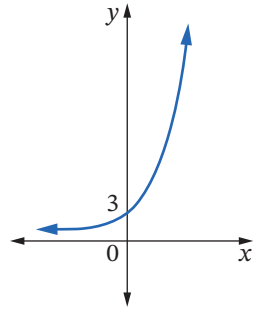
b How are the above graphs similar or different?

9 Which graph represents $y = 3(4^x)$? Select **A**, **B**, **C** or **D**. R C

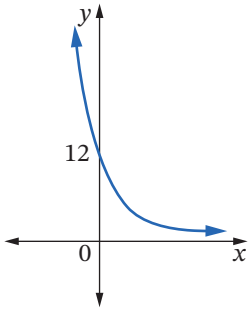
A



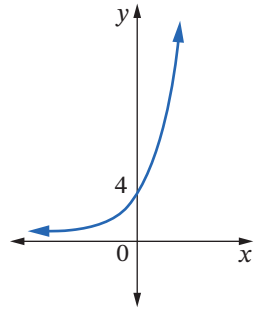
B



C



D



10 **a** Sketch each exponential curve, showing the y-intercept and asymptote. R C

i $y = 4(3^x) + 2$ **ii** $y = 4(3^x) - 1$

b How are the 2 graphs similar?

c How are the 2 graphs different?

7.12

PATH

EXAMPLE 21

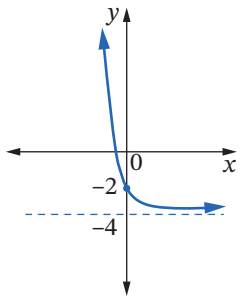
EXAMPLE 22

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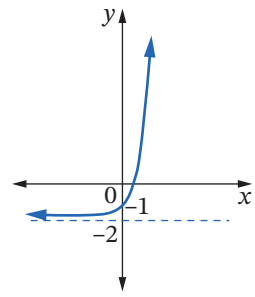
PATH

11 Which graph represents $y = 2(3^x) - 4$? Select **A**, **B**, **C** or **D**. R C

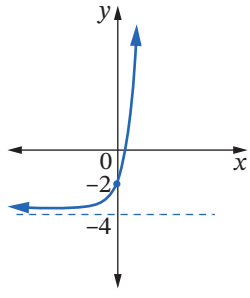
A



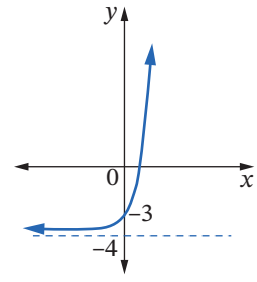
B



C



D

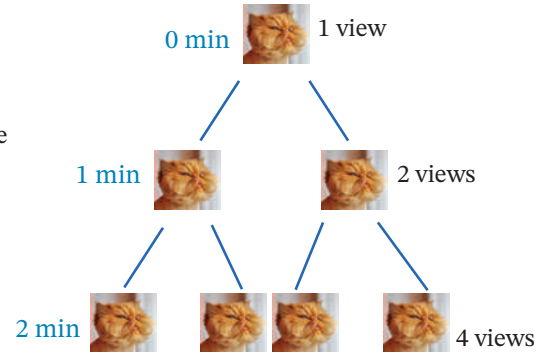


Investigation

Videos going viral

Use a spreadsheet to help you with this investigation.

With the invention of social media and social networks, we can now communicate with each other instantly. However, what often starts off as a simple message between friends or an online video post can quickly multiply at an alarming rate, be seen by the general public worldwide and 'go viral' (spread like a virus).



Zanna Pesnina/Shutterstock.com

Suppose a video of an angry cat is shared with a friend who one minute later shares it with 2 other friends. This occurs every minute so that every minute the number of *new* views doubles. This is an example of something growing exponentially, according to the formula $y = 2^x$, where x represents number of minutes.

1 Use a spreadsheet to calculate the number of new views after:

a 15 min **b** 30 min **c** 1 hour **d** 2 hours
e 4 hours **f** 10 hours **g** 20 hours **h** 30 hours

2 Find how long it will take until the total number of new views reaches:

a 64 **b** 256 **c** over 500 **d** over 1000
e over 3000 **f** over 10 000 **g** over 1 million **h** over 4 million

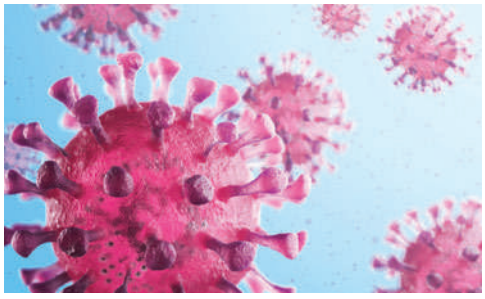
3 Use the spreadsheet's **Graph Option** to graph the viral pattern. Describe the shape of the graph.

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Did you know?

Exponential growth



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When an increase can be described using an exponential equation, it is called exponential growth. Examples include the growth of a virus, population (people or animals) and compound interest investments.

Population growth is monitored in different countries through the fertility (birth) and mortality (death) rates as well as migration. The data collected for these figures can often be modelled as an exponential function. By modelling the changes in population, predictions of future changes in population can be simulated and towns and cities can prepare for possible expansion in the numbers of schools, hospitals, housing and other necessary infrastructure.

At what rate is the population of Australia growing?
What about the world's population?

Exponential growth and decay 7.13

Any quantity that increases according to the exponential equation $y = k(a^x)$, where $a > 1$, is showing **exponential growth**, increasing slowly at first and then more quickly.

Examples of exponential growth are population growth, an investment increasing with compound interest, and the number of people or computers affected by a virus.

Any quantity that decreases according to the exponential equation $y = k(a^{-x})$, where $a > 1$, is showing **exponential decay**, decreasing quickly at first and then more slowly.

Examples of exponential decay are the cooling of a hot liquid, the depreciating value of an asset and radioactive material decay.

We have studied compound interest and depreciation in Chapter 1, *Interest and depreciation*.

Example 23

The future value $\$FV$ of an amount of \$80, invested at 5% p.a. interest compounded annually, after n years is given by the formula $FV = 80(1.05)^n$.

a Complete this table of values, correct to the nearest dollar.

n	0	10	20	30	40	50
FV						

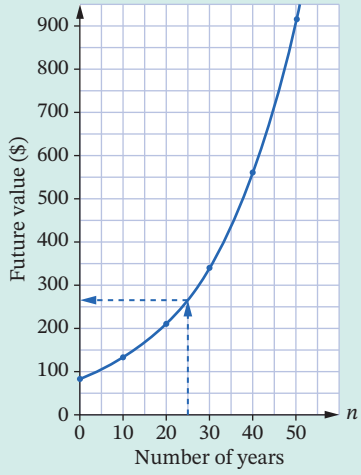
b Graph the future value formula and use the graph to find the future value of the investment after 25 years.

Solution

a

n	0	10	20	30	40	50
FV	80	130	212	346	563	917

b



Using the graph, the value of the investment after 25 years is about $270.

(Using the formula, the value of the investment is $270.91)

Example 24

The value $\$V$ of a car which depreciates annually is given by the equation $V = 50\,000\,(1.25)^{-n}$, where n is the number of years.

a Graph the equation for values of n from 0 to 5.

b What is the vertical intercept and what does it represent?

c Use the graph to estimate the value of the car after 3.5 years.

d Use the equation to determine when the car's value will drop to below $10 000.

Solution

a Completing a table of values with V rounded to the nearest $100, using the formula.

n	0	1	2	3	4	5
V	50 000	40 000	32 000	25 600	20 500	16 400

b The vertical intercept is 50 000, which is the original value of the car.

c From the graph, the value of the car after 3.5 years is $23 000.

(Using the formula, the value of V is $22 897)

d Use guess-and-check:

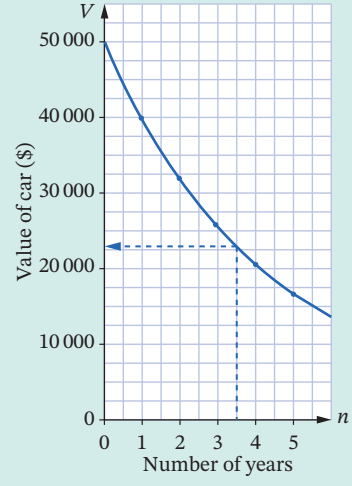
$n = 5$, $V = 16\,400$

$n = 6$, $V = 13\,107$

$n = 7$, $V = 10\,485$

$n = 8$, $V = 8388$

The value of the car will be less than $10 000 after 8 years.



EXERCISE 7.13 ANSWERS ON P. 645

Exponential growth and decay U F R C

EXAMPLE 23

7.13

1 The population of a small town grows according to the equation $P = 1200\,(1.095)^n$, where P is the population and n is the number of years after the end of 2025. R C

a Copy and complete the table of values correct to the nearest 10.

n	0	1	2	3	4	5	6	7	8	9	10
P											

b Graph the exponential equation for n from 0 to 10. Use a scale of 1 cm = 2 years on the horizontal axis and a scale of 1 cm = 400 people on the vertical axis.

c What is the vertical intercept and what does it represent?

d Use a guess-and-check method to find in what year the population will be triple the original population.

2 Bacteria cells grow by continually dividing into 2 cells. So 1 cell will divide into 2 cells, the 2 cells divide into 4 cells and so on. The number of bacteria cells (B) present can be found using the exponential equation $B = 10(2)^n$, where n is the number of times the cells divide. The cells divide every 20 minutes. R C

a Copy and complete this table of values and then graph the equation.
Use a scale of 1 cm = 100 bacteria on the vertical axis.

n	0	1	2	3	4	5	6
B							

b How many bacteria were there initially?

c How many bacteria were present after:

i 2 hours? **ii** 3 hours? **iii** 5 hours?

3 The future value of \$1500 invested at 6.5% p.a. interest compounded annually for n years is given by the equation $FV = 1500(1.065)^n$. R C

a Copy and complete this table correct to the nearest dollar.

n	0	2	4	6	8	10
FV						

b Graph the equation using a scale of 1 cm = 2 years on the horizontal axis and 1 cm = \$200 on the vertical axis

c What is the vertical intercept and what does it represent?

d Use your graph to find the what the future value is after:

i 3 years **ii** 7 years

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EXAMPLE 24

4 A ball is dropped from a window and the height (H cm) it reaches each bounce (x) is given by the equation $H = 400(1.55)^{-x}$. R C

a Copy and complete this table of values (to the nearest whole number).

x	0	2	4	6	8	10
H						

b Graph the equation using a scale of 1 cm = 50 cm on the vertical axis.

c What is the vertical intercept and what does it represent?

d Use the graph to find the height of the ball on the 5th bounce.

e After which bounce will the height be less than 20 cm?

f Use the equation to find the height of the ball on the 14th bounce.

5 The value of a car that depreciates annually is given by the equation $V = 35\,000(1.111)^{-n}$, where n is the number of years. R C

a Graph the equation for values of n from 0 to 8.

b What is the vertical intercept and what does it represent?

c Use the graph to estimate the value of the car after 3.5 years.

d Use the equation to determine when the car's value will drop to below $10 000.

6 The half-life of caffeine in the human body is about 5 hours. This means that if you consume 20 mg of caffeine, then after 5 hours you will still have 10 mg in your body.

An equation for the amount of caffeine (C mg) left in the human body t hours after drinking a cup of coffee (about 230 mL) is $C = 95(2)^{-0.2t}$. R C

a Copy and complete this table of values and graph the equation.

t	0	5	10	15	20	25
C						

b How much caffeine was originally in the cup of coffee?

c Use the graph to find how much caffeine will be in the human body after:

i 7 hours **ii** 16 hours **iii** 24 hours

d After how many hours will the amount of caffeine be less than 20 mg?

7 House prices ($P \times \$100\,000$) in a city are increasing according to the equation $P = P_0(1.15)^n$ where $P_0 \times \$100\,000$ is the original price of a house and n is the number of years after the end of 2025. Note that P and P_0 are values multiplied by $100 000. R C

a The original price of a house is $500 000.
Copy and complete this table, correct to one decimal place.

n	0	1	2	3	4	6
P (in $100 000s)	5					

b Draw the graph of the equation.

c What is the vertical intercept and what does it represent?

d Use the graph to find the value of the house at the end of 2030.

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8 A certain radioactive substance decays such that its mass (M kg) after t years is given by the equation $M = 100(2.718)^{-0.016t}$. R C

a Copy and complete this table of values correct to one decimal place.

t	0	10	20	30	40	60
M (kg)						

b What was the original mass of the radioactive substance?

c Draw the graph of the exponential equation. Use a scale of 1 cm = 10 years on the horizontal axis and 1 cm = 10 kg on the vertical axis.

d Use your graph to find the mass remaining after 25 years.

e The half-life of the substance is the time it takes to decay to half of the original mass. Use the graph to determine the half-life of the substance.

f Use the equation to find correct to 2 decimal places what amount of the substance remains after:

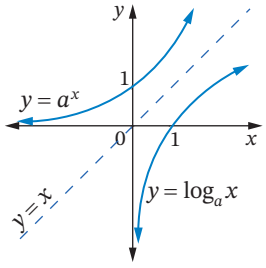
i 100 years **ii** 200 years **iii** 300 years

Graphing logarithmic equations $y = \log_a x$

7.14

PATH

Remember from Chapter 6, *Equations, inequalities and logarithms*, that the **logarithm** of a number is the power of the number, to a given positive base. For example, the logarithm of 32 to the base 2 is 5, written $\log_2 32 = 5$, because $2^5 = 32$. This means that $y = \log_2 x$ is the inverse or opposite equation to $y = 2^x$. If you make x the subject of $y = \log_a x$, then you get $x = a^y$. The graphs of $y = a^x$ and $y = \log_a x$ have symmetry. The diagram shows that one is a reflection of the other in the dotted line $y = x$.



Note that as x increases, the graph of $y = \log_a x$ increases slowly because y is a *power*. For example, since $\log_2 16 = 4$, this means that (16, 4) is a point on the graph of $y = \log_2 x$ (and (4, 16) is a point on the graph of $y = 2^x$).

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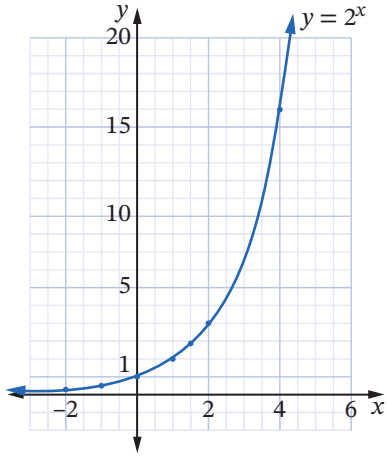
The table of values and graph for $y = 2^x$ and $y = \log_2 x$ are shown below for comparison.

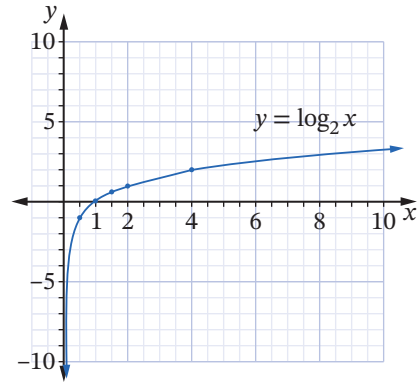
$y = 2^x$

x	−2	−1	0	$\frac{1}{2}$	1	$1\frac{1}{2}$	2	4
y	$\frac{1}{4}$	$\frac{1}{2}$	1	1.4	2	2.8	4	16

$y = \log_2 x$

x	−2	−1	0	$\frac{1}{2}$	1	$1\frac{1}{2}$	2	4
y	–	–	–	−1	0	0.6	1	2





The logarithmic curve $y = \log_a x$ (where $a > 1$)

- The curve is always increasing, but slowly with a near-flat gradient
- The y-axis is an asymptote
- The x-intercept is always 1 since $\log_a 1 = 0$
- The curve is always on the right side of the y-axis because logarithms are only meaningful for positive numbers.

EXERCISE 7.14 ANSWERS ON P. 647

Graphing logarithmic equations $y = \log_a x$ U F R C

Graphing technology can be used in this exercise.

1 Describe the shape of the graph of $y = 2^x$ and $y = \log_2 x$.
What are their x- and y-intercepts and asymptotes? R C

2 On the same set of axes, graph the functions $y = 3^x$ and $y = 4^x$. Draw the reflections of each graph in the line $y = x$ to illustrate $y = \log_3 x$ and $y = \log_4 x$.
Describe the differences and similarities between the graphs:

a $y = 2^x$, $y = 3^x$ and $y = 4^x$ **b** $y = \log_2 x$, $y = \log_3 x$ and $y = \log_4 x$

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3 Which is larger, 2^x, 3^x or 4^x, when: R C

a $x < 0$? **b** $x > 0$?

4 Which is larger, $\log_2 x$, $\log_3 x$ or $\log_4 x$, when: R C

a $x < 1$? **b** $x > 1$?

5 The log key on a scientific calculator is used to find $\log_{10} x$, where $x > 0$.
Evaluate each logarithm correct to 4 decimal places.

a $\log_{10} 20$ **b** $\log_{10} 627$ **c** $\log_{10} 5426$

d $\log_{10} 8.2$ **e** $\log_{10} 2.5$ **f** $\log_{10} 0.76$

6 Graph $y = \log_{10} x$ for values of x from 0 to 10.

7 Use the graph of $y = \log_{10} x$ to estimate: R

a $\log_{10} 3$ **b** $\log_{10} 6.5$ **c** $\log_{10} 8.4$ **d** $\log_{10} 0.5$

Compare your answers with those found using your calculator.

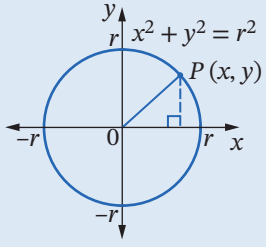
PATH

Graphing circles $(x - a)^2 + (y - b)^2 = r^2$ 7.15

The equation of a circle, centre (0, 0)

The equation of a circle with centre (0, 0) and radius r units is

$$x^2 + y^2 = r^2$$



PATH

Puzzle
Circle equations

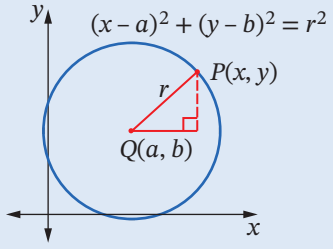
For example, the equation of a circle with centre (0, 0) and radius 7 units is $x^2 + y^2 = 49$.

If the centre of the circle is shifted from (0, 0) to new coordinates (a, b), then the equation of the circle changes from $x^2 + y^2 = r^2$ to $(x - a)^2 + (y - b)^2 = r^2$.

The equation of a circle, centre (a, b)

The equation of a circle with centre (a, b) and radius r units is

$$(x - a)^2 + (y - b)^2 = r^2.$$



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Proof

Let $P(x, y)$ be any point on the circle and $Q(a, b)$ be its centre so that the distance $PQ = r$ units.
Using the distance formula:

$$d = \sqrt{(x-a)^2 + (y-b)^2} = r$$
$$(x-a)^2 + (y-b)^2 = r^2$$

Example 25

Find the centre and radius of the circle with equation:

a $(x-2)^2 + (y-5)^2 = 9$ **b** $(x+3)^2 + (y+1)^2 = 16$

c $x^2 + (y-6)^2 = 1$ **d** $(x+1)^2 + (y-4)^2 = 40$

Solution

a Centre is (2, 5), radius $= \sqrt{9} = 3$ units

b Centre is (–3, –1), radius $= \sqrt{16} = 4$ units

c Centre is (0, 6), radius $= \sqrt{1} = 1$ unit

d Centre is (–1, 4), radius $= \sqrt{40} = 2\sqrt{10}$ units

Completing the square

$(x-2)^2 + (y-5)^2 = 9$ can be expanded to become $x^2 - 4x + y^2 - 10y + 20 = 0$

This is called the **general form of the equation of a circle.**

To find the centre and radius of a circle written in the general form, we must 'complete the square' on both x and y to arrange it in the form $(x-a)^2 + (y-b)^2 = r^2$.

The method of completing the square depends on these formulas for **perfect squares** studied in Chapter 5, *Products, factors and surds*:

$$(x+a)^2 = x^2 + 2ax + a^2$$
$$(x-a)^2 = x^2 - 2ax + a^2$$

Note that the last term, a^2, is the square of 'half the coefficient of x'.

Example 26

Find the numbers that complete the square in each equation.

a $x^2 + 10x + ___ = (x + ___)^2$ **b** $x^2 - 14x + ___ = (x - ___)^2$

Solution

a The coefficient of x is 10.
$\frac{1}{2}$ of 10 is 5, and $5^2 = 25$.
∴ The perfect square is
$x^2 + 10x + 25 = (x+5)^2$

b The coefficient of x is –14.
$\frac{1}{2}$ of –14 is –7, and $(-7)^2 = 49$.
∴ The perfect square is
$x^2 - 14x + 49 = (x-7)^2$

Example 27

Graph the circle with equation $x^2 + 4x + y^2 - 6y + 4 = 0$.

Solution

We need to rewrite the general equation into the form $(x - a)^2 + (y - b)^2 = r^2$.

$$x^2 + 4x + y^2 - 6y + 4 = 0$$

$$x^2 + 4x + y^2 - 6y = -4$$

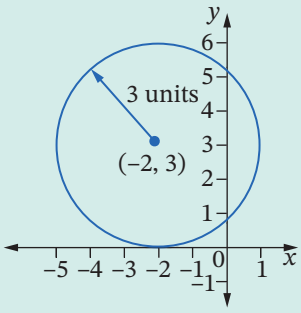
Moving the constant, 4, to the RHS

$$(x^2 + 4x + 4) + (y^2 - 6y + 9) = -4 + 4 + 9$$

Complete the square on $x^2 + 4x$ and on $y^2 - 6y$ and adding the same values to the RHS.

$$(x + 2)^2 + (y - 3)^2 = 9$$

This is the equation of a circle with centre $(-2, 3)$ and radius $\sqrt{9} = 3$ units.



EXERCISE 7.15 ANSWERS ON P. 647

Graphing circles $(x - a)^2 + (y - b)^2 = r^2$ U F R C

Some of this exercise may also be completed using graphing technology.

1 Find the centre and radius of the circle with equation:

a $x^2 + y^2 = 4$ **b** $x^2 + y^2 = 36$ **c** $x^2 + y^2 = 64$

d $x^2 + y^2 = 100$ **e** $x^2 + y^2 = 81$ **f** $2x^2 + 2y^2 = 50$

2 What is the equation of a circle with centre (0, 0) and radius 3 units? Select the correct answer **A**, **B**, **C** or **D**.

A $x^2 + y^2 = -9$ **B** $x^2 + y^2 = 3$ **C** $x^2 + y^2 = -3$ **D** $x^2 + y^2 = 9$

3 **a** Show that the point (8, 6) lies on the circle $x^2 + y^2 = 100$. R

b Show that the point (5, 9) does not lie on the circle $x^2 + y^2 = 100$.

c Does (5, 9) lie inside or outside this circle?

4 Given the equation of the circle $x^2 + y^2 = 4$, substitute each of the following points into the equation and determine whether the points are *inside*, *on* or *outside* the circle: R

a (0, 0) **b** (2, 0) **c** (3, 1) **d** (1, 1) **e** (−5, 2)

Foundation Standard Complex

PATH

EXAMPLE 25

5 Find the centre and radius of the circle with equation:

a $(x+2)^2+(y-4)^2=49$ **b** $(x-3)^2+(y-1)^2=1$

c $(x-9)^2+(y-12)^2=225$ **d** $x^2+(y+3)^2=4$

e $(x+6)^2+(y+1)^2=10$ **f** $(x+5)^2+(y+8)^2=16$

g $x^2+y^2=72$ **h** $(x+2)^2+(y-1)^2=50$

i $(x-2)^2+y^2=64$ **j** $(x-4)^2+(y+3)^2=\frac{25}{4}$

k $(x-3)^2+(y-4)^2=81$ **l** $3x^2+3(y+1)^2=\frac{1}{3}$

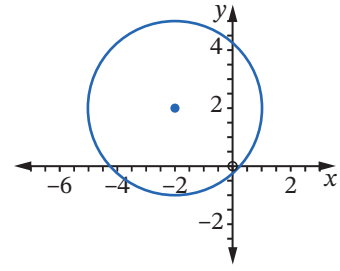
6 Find the equation of this circle. Select **A**, **B**, **C** or **D**.

A $(x-2)^2+(y+2)^2=3$

B $(x+2)^2+(y-2)^2=9$

C $(x+2)^2+(y-2)^2=4$

D $(x-2)^2+(y+2)^2=16$



7 Find the equation of the circle whose centre and radius are respectively:

a $(1, -2), 3$ **b** $(10, -11), 2$ **c** $(-3, 2), 10$

d $(0, -1), 1$ **e** $(-6, 2), \sqrt{5}$ **f** $(-1, 5), 2\sqrt{2}$

8 Graph each circle.

a $(x+1)^2+(y+1)^2=1$ **b** $x^2+(y-4)^2=25$

c $(x-1)^2+y^2=16$ **d** $(x+5)^2+(y+2)^2=4$

EXAMPLE 26

9 Find the numbers that complete the square in each equation. R C

a $x^2+2x+_____=(x+_____)^2$ **b** $p^2-6p+_____=(p-_____)^2$

c $m^2-8m+_____=(m-_____)^2$ **d** $k^2+4k+_____=(k+_____)^2$

EXAMPLE 27

10 Find the centre and radius of the circle given with equation: R C

a $x^2+6x+y^2-2y=15$ **b** $x^2-8x+y^2-4y=29$

c $x^2+4x+y^2-10y=7$ **d** $x^2+20x+y^2-12y+135=0$

e $x^2+y^2=4x-8y+5$ **f** $x^2-12x+y^2+6y+29=0$

g $x^2+y^2=20y-6x-28$ **h** $x^2+y^2+5=8x-2y$

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Identifying graphs 7.16

Graphs

Straight line:	$y = mx + c,\ ax + by + c = 0$
Parabola:	$y = ax^2 + c$
Exponential curve:	$y = a^x + c$
Exponential curve:	$y = ka^x + c$
Logarithmic curve:	$y = \log_a x$
Parabola:	$y = a(x - b)^2 + c$
Cubic curve:	$y = ax^3 + c$
Power curves:	$y = ax^n + c, y = a(x - b)^n$
Hyperbola:	$y = \frac{k}{x} + c,\ y = \frac{k}{x - b}$
Circle:	$x^2 + y^2 = r^2$
Circle:	$(x - a)^2 + (y - b)^2 = r^2$

PATH

Puzzles Matching graphs

Matching graphs (Advanced)

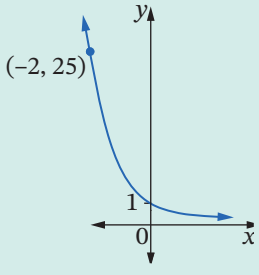
When matching graphs with equations, the x value and y value of a point on the graph may need to be substituted into the equation to verify that the equation represents the graph.

Video Identifying graphs

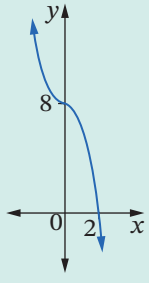
Example 28

Match each graph to its equation.

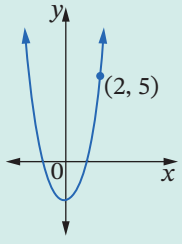
a



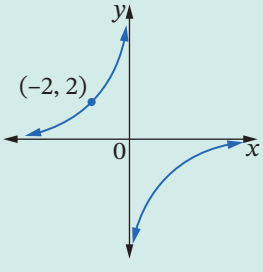
b



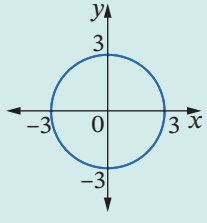
c



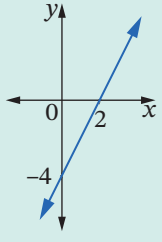
d



e



f



A $y = 2x - 4$

B $x^2 + y^2 = 9$

C $y = -\frac{4}{x}$

D $y = 2x^2 - 3$

E $y = 5^{-x}$

F $y = -x^3 + 8$

Solution

a An exponential curve that matches with E, $y = 5^{-x}$

Test point: (–2, 25)

LHS = 25

RHS = $5^{-(-2)} = 5^2 = 25$ = LHS

PATH

b A decreasing cubic curve with y-intercept 8 that matches with F, $y = -x^3 + 8$

Test point: (2, 0)

LHS = 0

RHS = $-2^3 + 8 = -8 + 8 = 0$ = LHS

c A concave-up parabola that matches with D, $y = 2x^2 - 3$

Test point: (2, 5)

LHS = 5

RHS = $2 \times 2^2 - 3 = 5$ = LHS

PATH

d A 'negative' hyperbola that matches with C, $y = -\frac{4}{x}$

Test point: (–2, 2)

LHS = 2

RHS = $y = -\frac{4}{-2} = 2$ = LHS

e A circle with centre (0, 0) and radius 3 that matches with B, $x^2 + y^2 = 9$

f A straight line with gradient 2 and y-intercept –4 that matches with A, $y = 2x - 4$

EXERCISE 7.16 ANSWERS ON P. 648

Identifying graphs U F R C

1 For each equation state whether its graph is a straight line (L), a parabola (P), an exponential curve (E) or a circle (C). R C

a	$y = 9x^2 - 4$	**b**	$y = 9x$	**c**	$y = 9^x$	**d**	$y = 9$
e	$x^2 + y^2 = 81$	**f**	$y = 3x - 8$	**g**	$y = 3x^2 - 8$	**h**	$y = 2x + 5$
i	$y = -x^2 + 6$	**j**	$y = 10^{-x}$	**k**	$y = 7x^2 + 2$	**l**	$x^2 + y^2 = 36$

EXAMPLE 28

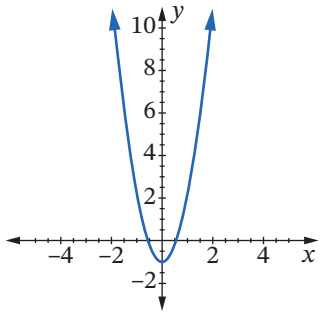
2 Match each equation with its graph. R C

a	$x = 4$	**b**	$y = 1 - x^2$	**c**	$y = 5$
d	$y = 3x^2 - 1$	**e**	$y = 3^x$	**f**	$x^2 + y^2 = 9$
g	$y = 3^{-x}$	**h**	$y = 2x^2$	**i**	$y = 9x^2 - 4$

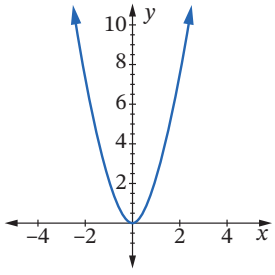
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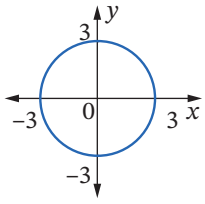
A



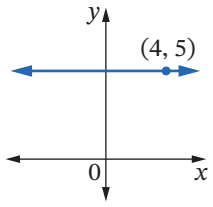
B



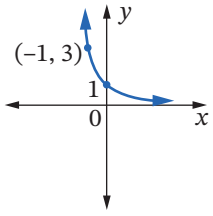
C



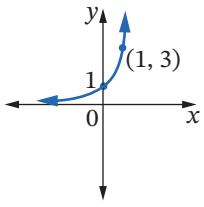
D



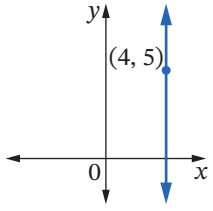
E



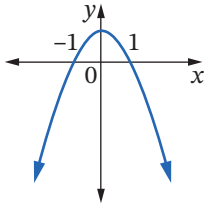
F



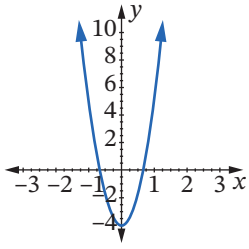
G



H



I



3 Sketch each equation, showing a point on the curve.

a $y = x^2 - 3$ **b** $y = 5^x$ **c** $y = -x^2 + 4$ **d** $x^2 + y^2 = 49$

e $y = \frac{3}{x}$ **f** $y = -2x + 4$ **g** $y = 3(2^{-x}) - 2$ **h** $y = x^3 - 8$

4 Find the y-intercept of the graph of each equation.

a $y = 3^x$ **b** $y = 2x^2 + 3$ **c** $y = 4(x + 1)^2 - 5$ **d** $y = \frac{4}{x+1}$

5 For each equation, state whether its graph is a parabola (P), a cubic (Q), a hyperbola (H), an exponential (E) or a circle (C). R C

a $y = \frac{9}{x}$ **b** $(x - 2)^2 + y^2 = 4$ **c** $y = 2(x - 2)^2$

d $y = \frac{-2}{x-4}$ **e** $y = 5(3^{-x}) - 4$ **f** $(x + 4)^2 + (y - 1)^2 = 15$

g $y = 4^x + 2$ **h** $y = -\frac{8}{x} - 1$ **i** $y = -\frac{1}{2}x^3 - 4$

PATH

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PATH

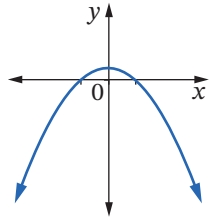
6 Match each equation with its graph. R C

a $y = -\frac{4}{x}$ **b** $y = (x-3)^3$ **c** $y = -x^3 + 1$ **d** $y = 1 - x^2$

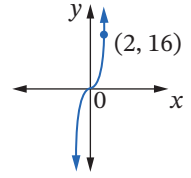
e $y = 2(3^x) - 5$ **f** $y = 2x^3$ **g** $y = \frac{4}{x} - 2$ **h** $y = (x+4)^2$

i $y = \log_2 x$

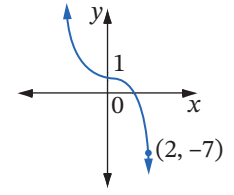
A



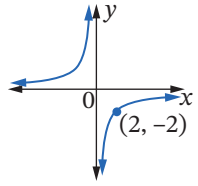
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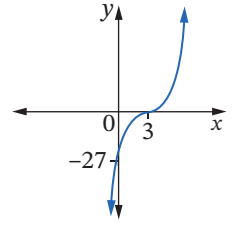
C



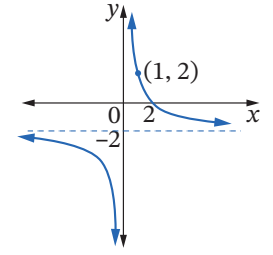
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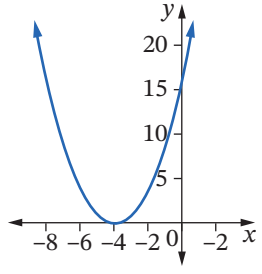
E



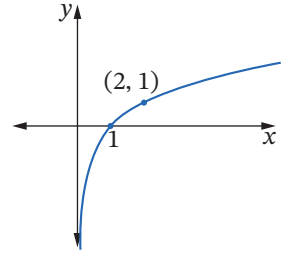
F



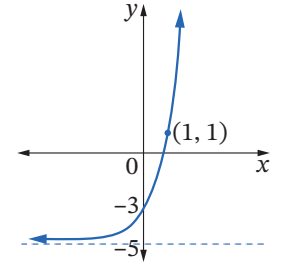
G



H



I



7 For each equation: R C

i write the name of its graph

ii find its y-intercept

iii find the equation of its asymptote(s)

a $y = 2^x$ **b** $y = 5^x + 1$ **c** $y = 4(10^x) - 3$

d $y = \frac{10}{x}$ **e** $y = \frac{2}{x-3}$ **f** $y = -\frac{1}{x} + 2$

8 Sketch the graph of each equation, showing all main features. R C

a $y = \frac{6}{x}$ **b** $y = 3^x + 2$ **c** $y = x^3 + 3$

d $y = 2(x-5)^2$ **e** $y = -\frac{1}{x+4}$ **f** $(x+5)^2 + (y-5)^2 = 4$

g $y = 8(2^{-x}) - 3$ **h** $y = 4 - (x-3)^2$

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Technology

Points of intersection

In this activity we will use graphing technology to find the point of intersection of a line and a curve.

1 Graph these equations of a line and parabola on the same screen.

$y = 3x - 4$ and $y = 2x^2 + x - 16$

2 How many points of intersection are there between the line and the parabola?

3 Click on each point of intersection to display its coordinates.

4 Graph each pair of equations to find the points of intersection of the line:

a $y = -x + 8$ and parabola $y = x^2 - 8x + 8$

b $x + y = 4$ and the exponential curve $y = 2^x$

c $y = 6x - 10$ and the parabola $y = 9x - x^2$

d $y = x - 3$ and hyperbola $y = -\frac{2}{x}$

e $y = x + 6$ and parabola $y = 5x^2$

f $15x + 8y = 17$ and exponential curve $y = 4^{-x}$

g $x - 2y = -6$ and hyperbola $y = \frac{8}{x}$

h $y = 2x - 1$ and hyperbola $y = \frac{15}{x}$

i $2x + y = -1$ and parabola $y = x^2 - 2x - 5$

j $x + 6y = 19$ and exponential curve $y = 3^x$

k $y = 2x - 5$ and circle $x^2 + y^2 = 10$

l $y = 4x$ and cubic curve $y = x^3$

m $x + y = 1$ and circle $x^2 + y^2 = 5$

n $y = 1 - 3x$ and cubic curve $y = 2 - x^3$

o $2x + y = 3$ and circle $(x - 2)^2 + (y + 1)^2 = 4$

p $y = 2 - 3x$ and cubic curve $y = \frac{1}{2}x^3 - 4$

Power plus ANSWERS ON P. 649

1 Graph the equation $y = \frac{1}{x-1} + 2$, showing all intercepts and asymptotes.

2 Sketch the graph of each equation and find the centre and radius of the semicircle.

a $y = \sqrt{16 - x^2}$ **b** $y = \sqrt{25 - x^2}$ **c** $y = -\sqrt{9 - x^2}$

CHAPTER 7 REVIEW

Quiz
Language of maths 7

Puzzle
Graphing curves crossword

Language of maths

asymptote	axis	centre	circle
coefficient	concave down	concave up	constant
conversion graph	direct variation	exponential decay	exponential growth
hyperbola	inverse variation	logarithmic	parabola
proportional	quadratic	radius	variable
vertex	vertical intercept	x-intercept	y-intercept

1 What is the coefficient of x^2 in the quadratic equation $y = 3x^2 + 10$?

2 What is the graph of a **quadratic equation** called?

3 True or false: The exponential curve $y = 2^x$ passes through the point $(0, 1)$.

4 In the variation equation $y = \frac{k}{x}$, which is the **constant of variation**?

5 Write down the equation of a parabola that is concave down and has a y-intercept of 3.

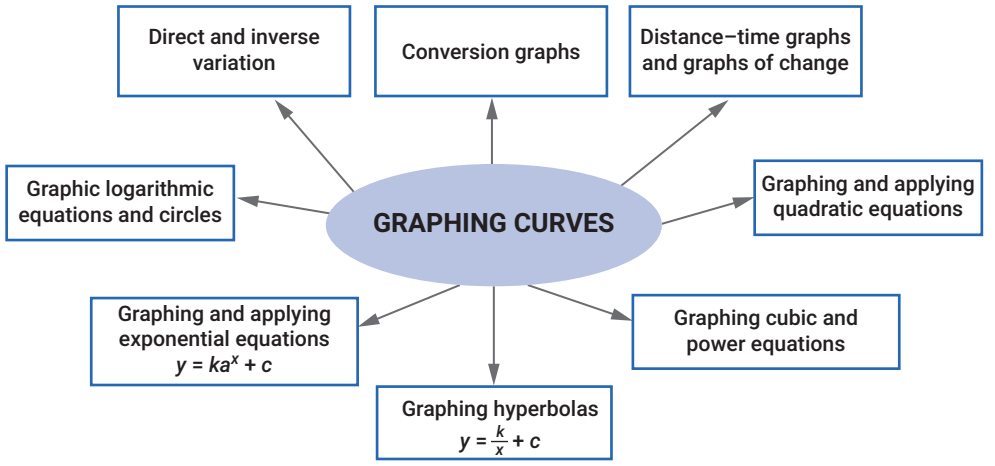
6 What is the asymptote of the exponential curve $y = a^x$?

Topic summary

- Which parts of this chapter were new to you?
- What is the difference between direct and inverse variation?
- Do you know the equations of a parabola, cubic curve, exponential curve, logarithmic curve and circle, and how to graph them?
- Explain how the graph of $y = 2x^2 + 3$ is different from the graph of $y = -2x^2 + 3$. How are they similar?

Worksheet
Mind map: Graphing curves (Advanced)

Print (or copy) and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.



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TEST YOURSELF 7 ANSWERS ON P. 650

1 H is directly proportional to t. If when $t = 12$, $H = 138$, find H when $t = 27$.

2 Air temperature, T (in degrees Celsius), varies inversely as its height, h (in metres), above sea level. At 400 m above sea level, the temperature is 15°C.
What is the temperature at 600 m above sea level?

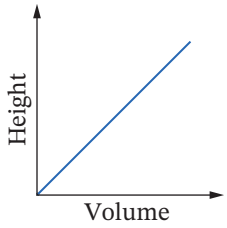
3 The graph in Example **5** on page 311 converts Australian dollars to UK pounds sterling. Use the graph to convert:

a $A70 to £ **b** £56 to Australian dollars

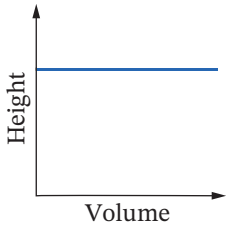
4 Draw a distance–time graph to show a person leaving home at an increasing speed and then slowing down.

5 The graphs below show the height of water in a bathtub over time. Match each graph to the correct description.

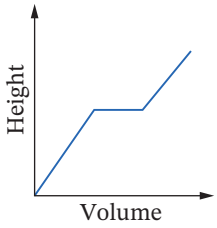
a



b



c



A The bath is almost full of water.

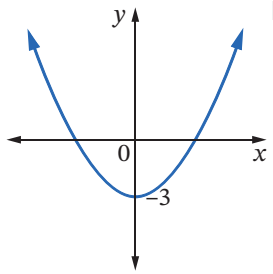
B Hot water is accidentally run into the bath, so the tap is turned off to let it cool. Then it is filled with cold water only.

C The bath is filled at a steady rate.

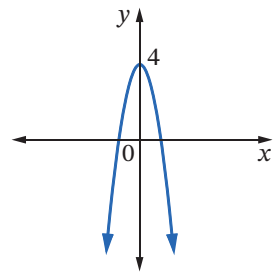
7.06

6 Match each graph to its correct equation.

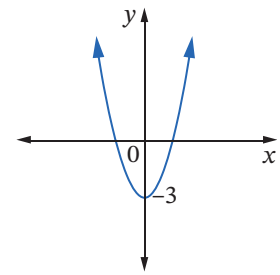
a



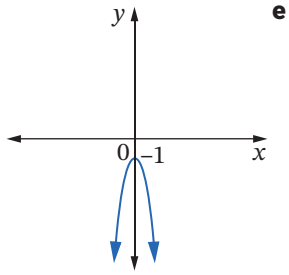
b



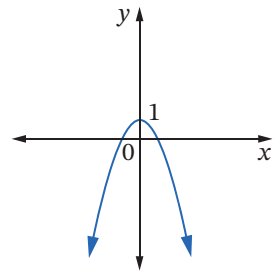
c



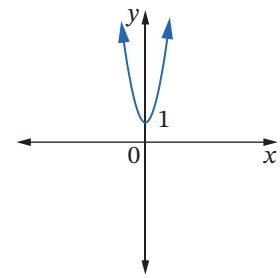
d



e



f



A $y = x^2 - 3$ **B** $y = 3x^2 + 1$ **C** $y = \frac{1}{2}x^2 - 3$

D $y = -x^2 + 1$ **E** $y = -4x^2 - 1$ **F** $y = 4 - 3x^2$

PATH

7.07

7 Graph the equation $y = 2(x + 3)^2$, showing all main features.

7.08

8 A stone is dropped from the top of a building and its height (h) in metres after t seconds is given by the quadratic equation $h = 70 - 4.9t^2$.

a Copy and complete this table of values and graph the equation.

Use a scale of 1 cm = 1 second on the horizontal axis and 1 cm = 10 m on the vertical axis.

Time (t seconds)	0	1	2	3	4
Height (h metres)					

b What is the vertical intercept and what does it represent?

c What is the height of the stone after 1.5 seconds?

d How far does the stone fall in the 3rd second?

e Use the graph to determine the time it takes for the stone hit the ground.

7.09

9 Graph the equation $y = x^3 + 2$, showing all main features.

7.10

10 Sketch the graphs of the equations $y = -3x^4$ and $y = -3(x - 2)^4$, showing all main features.

7.11

11 Sketch the graph of $y = \frac{2}{x+1}$, showing the x- and y-intercepts and a point on the curve.

7.12

12 Sketch each curve described below.

a $y = 4^x$ **b** $y = 4^{-x}$ **c** $y = -4^x$ **d** $y = -4^{-x}$

7.12

13 Sketch each curve below, showing the y-intercept and the asymptote.

a $y = 2(3^x) - 4$ **b** $y = 3(5^{-x}) + 2$

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14 Simone is given some money on her 5th birthday, and this amount is doubled each birthday until she turns 21. The amount of money ($\$A$) she receives each birthday is modelled by the exponential equation $A = 5(2^n)$, where n is the number of years after her 5th birthday. 7.13

a Copy and complete this table of values to graph the equation.

n (years)	0	1	2	3	4	6	8	10
A (cents)								

b How much money did Simone receive on her 5th birthday? How is this shown on your graph?

c How much did Simone receive on her 9th birthday (in dollars)?

d At what age will Simone receive more than $1500?

e How much will Simone receive on her 21st birthday?

15 The salvage value of a car ($\$V$) after n years can be calculated using the equation $V = 12\,000\,(1.24)^{-n}$. 7.13

a Copy and complete this table of values, correct to the nearest $10.

n	0	1	2	3	4	5
V						

b Use a scale of 1 cm = $1000 on the vertical axis to graph the equation.

c What is the vertical intercept and what does it represent?

d How much does the value of the car drop in the:

i 1st year? **ii** the 4th year?

e Use guess-and-check to find when the value of the car falls below $1000.

PATH

16 **a** Draw the graph of $y = 5^x$ and then draw its reflection in the line $y = x$. 7.14

b Write down the equation of the reflected graph.

17 Find the centre and radius of each circle described below. 7.15

a $x^2 + y^2 = 100$ **b** $x^2 + y^2 = 36$ **c** $x^2 + y^2 = 49$

d $(x + 5)^2 + (y - 6)^2 = 81$ **e** $(x - 10)^2 + y^2 = 225$ **f** $(x + 7)^2 + (y + 10)^2 = 80$

18 Find the centre and radius of the circle with equation $x^2 + 8x + y^2 - 4y = 29$. 7.15

TEST YOURSELF 7

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7.16

19 Match each equation with its correct graph.

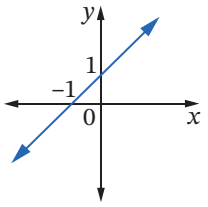
a $y = \frac{1}{4}x^2$ **b** $y = 3^x$ **c** $y = -2x^2 - 1$

d $x = -5$ **e** $y = -3x^2$ **f** $y = 2x^2 - 1$

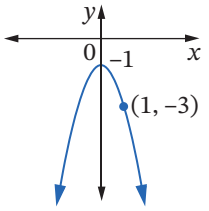
g $y = 3^{-x}$ **h** $y = -\frac{8}{x}$ **i** $y = x^2$

j $y = x + 1$ **k** $y = -5$ **l** $y = -2 - 2x$

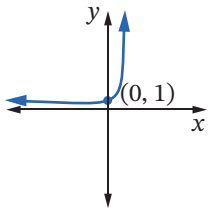
A



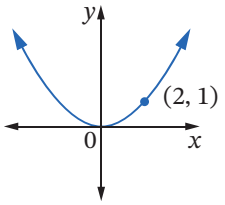
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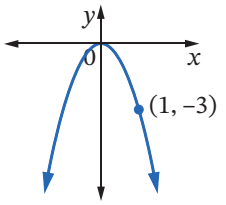
C



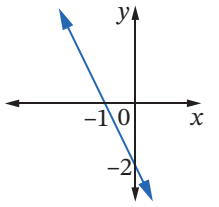
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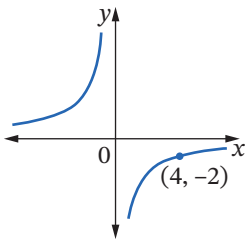
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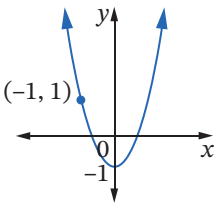
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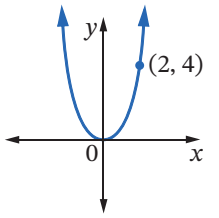
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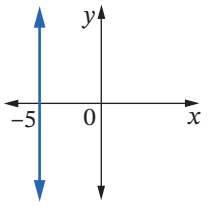
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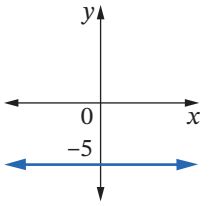
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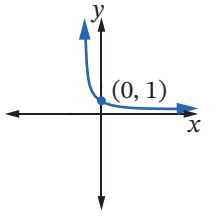
J



K



L



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